



Impact of crop management practices on maize yield: Insights from farming in tropical regions and predictive modeling using machine learning

Showkat Ahmad Bhat^{a,b,*}, Syed Asif Ahmad Qadri^c, Vijay Dubbey^c, Ishfaq Bashir Sofi^d, Nen-Fu Huang^{b,**}

^a Center of Innovative Incubator, National Tsing Hua University, Hsinchu, 1300044, Taiwan, ROC

^b Department of Computer Science, National Tsing Hua University, Hsinchu, 1300044, Taiwan, ROC

^c College of Electrical Engineering and Computer Science, National Tsing Hua University, Hsinchu, 1300044, Taiwan, ROC

^d Department of Electrical and Computer Engineering, Concordia University, Montreal, H3G1M8, Canada

ARTICLE INFO

Keywords:

Maize yield
Crop management
Machine learning methods
Tropical agriculture
Explainable artificial intelligence

ABSTRACT

Understanding the influence of crop management practices on maize yield is crucial amidst the changing environmental conditions. Smallholder farming in tropical regions has long puzzled decision-makers in terms of maize management. Balancing alternative management practices, environmental factors, and economic outcomes is essential. In this study, we examine the relationships between maize yield and various factors including climate variables, soil quality parameters, cultivars, tillage practices, and fertilizer usage in Chiapas, Mexico. Pearson's correlation coefficient and T-test were employed to determine the statistical significance of the correlations. Our findings reveal strong positive associations between maize yields and factors such as planting, cultivar, vapor pressure, temperature, solar radiations and fertilizer inputs for nitrogen, phosphorus, and potassium in lower elevation farms. Conversely, there is a negative correlation with elevation, slope, and system. Extensive data analysis was conducted to investigate the impact of different crop management practices on yield, utilizing both data visualization and analytics. Additionally, maize yield prediction was performed using six hybrid machine learning (ML) models, employing Grid Search and Random Search optimization techniques. Based on evaluation, Grid Search-based XGBoost exhibited the most accurate prediction results. Furthermore, explainable artificial intelligence (XAI) was employed to assess the individual impact of each input parameter on the ML model output.

1. Introduction

Maize (*Zea mays* L.) is a staple crop crucial for food assistance programs, livestock feeding due to its high starch content, and various industrial applications [1–4]. It ranks third among cereals globally but faces challenges with low and inconsistent yields. Despite its potential yield range of 6–13.9 Mg ha⁻¹ [5], actual yields remain below 5 tons ha⁻¹, with a significant yield gap between research stations and farmer fields [6]. Mexico, a top maize producer, averages 2.9 tons per hectare, much lower than the global average of 5.1 tons, and imports 34.12 % of its supply [2,7,8]. In sub-Saharan Africa, maize yields can increase significantly with proper fertilizer management, from 1.4 to 3.9 tons ha⁻¹, highlighting the potential for yield improvement.

The growing recognition of global climate change as a threat to

agriculture highlights its impact on maize (*Zea mays* L.), a crucial crop for global food security [9,10]. Rising temperatures and changing precipitation patterns have already negatively affected maize yields [11]. Maize, domesticated 8000 years ago in Mexico, is essential to local and global food security. Research has extensively examined factors such as water stress, temperature, solar radiation, tillage practices, fertilizers, soil types, and other environmental variables on maize yield [2,12]. Studies have explored varying irrigation levels, high-temperature stress using crop models, and rainfall variations, emphasizing the importance of understanding these factors to mitigate climate change's impact on maize production [13,14].

An investigation by Ref. [15] examined the impact of phosphorus (P) on maize yield and agronomic efficiency in China, using data from 7025 field trials (2005–2010). The study quantified the effects of soil

* Corresponding author. Center of Innovative Incubator, National Tsing Hua University, Hsinchu, 1300044, Taiwan, ROC.

** Corresponding author.

E-mail addresses: showkatbhat1994@gmail.com (S.A. Bhat), nfhuang@cs.nthu.edu.tw (N.-F. Huang).

<https://doi.org/10.1016/j.jafr.2024.101392>

Received 13 April 2024; Received in revised form 3 June 2024; Accepted 2 September 2024

Available online 4 September 2024

2666-1543/© 2024 Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

conditions and management practices on yield response to phosphorus (YRP) and agronomic efficiency of phosphorus (AEP). Another study [16] evaluated different maize genotype's grain yield, nitrogen (N) usage efficiency, rainwater usage efficiency, and nitrate-N residue, highlighting the importance of appropriate nitrogen management and genotype selection in rain-fed regions. Understanding macronutrient dose-response relationships is vital for improving maize yields. A study [17] explored the effects of nitrogen (N), phosphorus (P), potassium (K), and sulfur (S) application rates on maize yield and agronomic efficiency when other nutrients were applied in balance.

Nutrient management is crucial as soil fertility has been compromised by repeated cropping and insufficient nutrient replenishment. To promote higher yields, restoring soil fertility and implementing minimal tillage practices are recommended [18]. Smallholder farmers face challenges due to declining soil fertility, insufficient fertilizer use, continuous cropping, intensive tillage, and crop residue clearance, leading to soil degradation and erosion [19]. Researchers have developed methods to forecast agricultural productivity, which are essential for stakeholders in the agricultural food production chain, including farmers, business owners, and governments [20,21]. Accurate crop output forecasts support food security strategies, efficient crop management, and the operations of agri-food firms [22].

Remote sensing is widely used for predicting yields, leveraging various sensors and satellite missions to monitor agricultural yields throughout the growing season [23,24]. There are two main prediction techniques: growth model-based methods and statistical approaches. Growth models estimate field-level output well but struggle with regional output due to calibration challenges [20]. Statistical methods often use vegetation indices (VIs) or meteorological data combined with yield data, but linear regression can oversimplify the relationships between environmental factors and yield [15,25]. Growth model-based techniques perform well at estimating field-level output when agronomic practices are known, but they face limitations in estimating regional output due to the challenges of obtaining calibration factors for large-scale applications. On the other hand, statistical techniques often utilize vegetation indices (VIs) or meteorological data, or a combination of both, along with yield data. However, linear regression approaches often oversimplify the complex relationships between environmental factors and yield. To address this issue, machine learning (ML) algorithms have shown promising performance in yield estimation across different crops and conditions, making them an intriguing tool in this field [26]. In fact, previous studies have successfully employed various ML algorithms for yield forecast regression [20,27–29].

Mexican maize has consistently been reported to be negatively affected by climate change [30–32]. It is imperative to consider the impact of various parameters on maize production. While climate projections vary across different regions in Mexico, the prevailing trend of negative impacts, particularly on precipitation patterns, persists. Therefore, it becomes crucial to examine the role of other parameters, such as elevation, slope, clay content, planting (day of the year (DOY)), cation exchange capacity (CEC), soil organic matter (SOM) concentration, genotype, nitrogen (N), phosphorus (P), potassium (K), pH, precipitation (Prcp_T), solar radiation (Srad_T), temperature (Tmax_T and Tmin_T), vapor pressure (Vp_T), tillage practices, and system characteristics.

Previous studies have extensively explored the impact of individual factors such as soil pH, precipitation, solar radiation, and temperature on maize yield. However, these studies often did not account for the complex interactions between these factors, nor did they consistently consider the site-specific conditions that can significantly alter outcomes. Our study aims to address these limitations by providing a comprehensive analysis that integrates multiple environmental variables and management practices. By using advanced statistical techniques, we examine how factors such as field management practices, soil parameters, genotype, and ambient weather conditions collectively influence maize yield. This holistic approach allows us to identify key

drivers of yield variability and develop strategies for optimizing maize production under diverse conditions. The benefits of this research are twofold. Firstly, it provides actionable insights for farmers and agronomists to enhance maize yield through targeted interventions. Secondly, it contributes to the scientific understanding of crop-environment interactions, paving the way for future research in sustainable agriculture.

This research paper presents several contributions in understanding the impact of crop management practices on maize yield, specifically in smallholder farming in tropical regions, as outlined below:

- Comprehensive Understanding: Our study aims to gain a comprehensive understanding of the effects of each parameter on maize productivity and the interplay between these factors. By analyzing the correlations between maize yield and climate variables, soil quality parameters, cultivars, tillage practices, and fertilizer usage, we provide insights into their influence on maize production.
- Prediction for Optimized Crop Management: Through machine learning techniques, we assess the potential of model-generated predictions for optimizing crop management decisions. By evaluating the performance of different ML models, we offer insights into predictive modeling effectiveness for maize yield estimation.
- Knowledge Expansion: The findings of this study contribute to the existing knowledge base by shedding light on maize production factors. By exploring the spatial and temporal patterns of alternative management practices for maize and considering the influence of various parameters and operations, we expand the understanding of maize management practices.
- Informed Decision-Making: This research paper discusses the implications of our findings in terms of sustainable agriculture and provides recommendations for optimizing maize management in tropical regions. Our insights enable informed decision-making for smallholder farmers and stakeholders in the agricultural sector.
- Reference for Future Research: This study serves as a valuable reference for similar research endeavors in the emerging field of data-driven agriculture. It provides a detailed framework for the detailed analysis and understanding of maize management practices, specifically in Chiapas, Mexico. Our approach demonstrates how comprehensive data collection and advanced analytical methods can be applied in agricultural research to drive meaningful insights, which is a key aspect of modern data-driven agricultural studies.
- XAI for Interpreting Model Output: Using explainable artificial intelligence (XAI), we assess the individual impact of each input parameter on the ML model output. This enhances the interpretability of the predictive model, providing valuable insights into the importance of different crop management practices in determining maize yield.

By offering these contributions, our research not only advances the understanding of maize management practices but also addresses the need for data-driven decision-making in agricultural settings, ultimately promoting sustainable and efficient crop management practices in tropical regions.

2. Materials and methods

Our main objective is to comprehensively evaluate different maize crop management practices from agricultural, germplasm improvement, and environmental perspectives, while also assessing the performance of a hybrid machine learning (ML) framework in accurately predicting maize yields. This section specifically examines the dataset utilized, hyperparameter tuning, and the ML algorithms employed as illustrated by Fig. 1. It provides detailed insights into the prediction/forecasting framework, including the inputs fed into the models, the data processing tasks involved, the specific predictive models selected, and the evaluation metrics used to effectively compare performance of the various ML models.

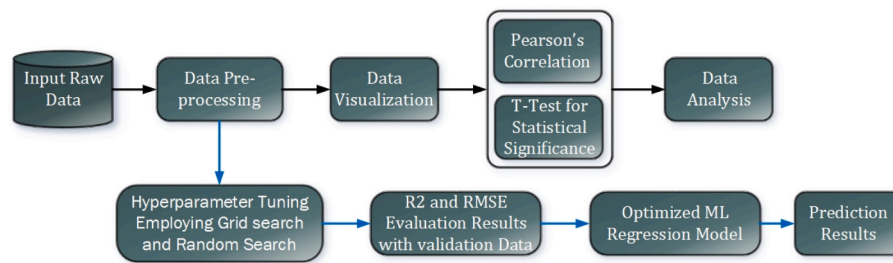


Fig. 1. Illustration of the methodology for maize yield prediction: Data pre-processing, visualization, analysis, hyperparameter tuning, and prediction.

2.1. Experimental design and data description

The data was collected from CIMMYT's knowledge hub in Chiapas, Mexico, where maize farming comprises two distinct systems: Landrace-based system, consisting of small-scale, low-input farmers who consume their grain, and the Hybrid system, involving medium-scale, medium-input, mechanized semi-commercial farmers who sell to local markets. Agronomic experimentation at the hub encompasses various levels: research platforms serve as controlled experiments; field demonstrations occur on farmer's land, facilitating side-by-side comparisons of different technologies and management practices; extension areas enable farmers to implement tested management changes after demonstrations; farmer impact areas are regions where innovations have been adopted, often without direct farmer involvement in experiments [33].

The dataset comprises data collected from the extension and module areas over seven growing seasons, spanning from 2012 to 2018. Field visits were conducted during this period to document information related to field identification, yields, and management practices. CIMMYT utilized the CIMMYT-designed system, known as Bitacora electronic MasAgro (BEM), for data entry. This system facilitated the input of farmers' information into field books created by CIMMYT. BEM supported logic, predefined answer ranges, data repetition within sub-structures and the geo-referencing of data. Data gatherers had the capability to submit information both offline and online, allowing for submissions at any time. The observations in the dataset pertain to individual crops and their associated practices in single fields, typically small in size (less than 5 ha) or specific portions of larger fields. Fig. 2 shows the geographical distribution of study sites across the Chiapas region. Each site is marked with a distinct color and "O" marker, uniformly distributed throughout the region to represent the diverse range of agricultural fields involved in our study. However, as the number of sites exceeds the number of unique colors in the colormap, colors repeat. Each point represents a distinct experimental site, differentiated by Field

ID and plotted according to its geographical coordinates (Longitude and Latitude). The map provides an overview of the spatial spread of the research. It underlines the extensive coverage of maize yield studied across different geographical locations within the region.

To identify each field, a unique ID, geographical coordinates (latitude and longitude with a precision of at least 7 digits), and elevation were automatically assigned. Additionally, farm advisors recorded farmer's fertilization doses and timings, including N, P2O5, and K2O doses. The dataset was collected from plots where five typical points were harvested to calculate grain yield. Each plot had dimensions of 5 by 1.5 m, equivalent to two rows of maize. For reporting purposes, the moisture content of the grain was standardized at 14 %. Due to the extended harvesting period, especially among subsistence farmers in the Chiapas regions of Ocosingo and Los Altos, yield estimation in extension areas relied on farmers' assessments. Supplementary data on average grain yields by municipality were obtained through SIAP surveys [34]. In the analysis, data integrity was ensured by eliminating extreme values, missing data points, and outliers falling outside the state's boundaries. For data manipulation, R software in conjunction with the dplyr package [35] were utilized.

To protect the identification of specific agricultural fields, spatial blurring was applied, reducing geographical coordinates to two decimal places. This measure ensured privacy while maintaining data accuracy. To enrich our analysis, polygon maps of soil units, complete with their functional attributes, were generated. These maps were constructed using soil samples, sourced from Mexico's National Institute of Statistics and Geography (INEGI). The process involved overlaying point coordinates from the field data onto the soil map polygons, a task facilitated by the R software package "sf" [36]. For a comprehensive soil analysis, key parameters including clay content, cation exchange capacity (CEC), soil organic matter (SOM), and pH values were tested. Furthermore, slope and elevation calculations were derived using digital elevation data obtained from the SRTM project [37].

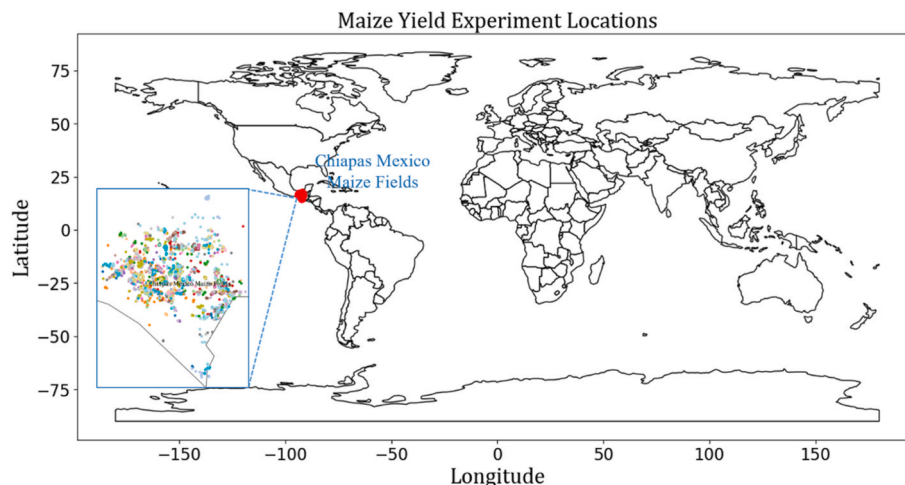


Fig. 2. Geospatial distribution of maize yield experiment sites in chiapas, Mexico.

Incorporating meteorological context into the research, weather datasets were compiled by leveraging Daymet gridded surface weather data, characterized by a spatial resolution of 1 km [38]. These datasets served as an invaluable source of meteorological information, integral to this study. To harness the potential of these datasets, R’s “raster” function were employed, superimposing field measurement point coordinates onto the Daymet gridded data. This allowed to extract essential values for various meteorological parameters, including precipitation, solar radiation, maximum and minimum temperatures, and vapor pressure, for each pixel. To streamline the modeling process and reduce the number of variables, transformed daily data into ten-day intervals. This enabled to group meteorological data based on planting dates, encompassing a period of 60 days before and up to 240 days after the initial planting. This approach resulted in the creation of 30 new characteristics for each variable, extending analysis beyond the typical growing season and capturing a broader range of environmental conditions.

The dataset used in this study consists of four main categories of information: field identification, maize yield and field management, soil parameters, and ambient weather conditions [39]. Over the course of seven growing seasons spanning from 2012 to 2018, farm advisers conducted routine field visits. They diligently recorded crucial information on field identification, maize production, and field management practices. The dataset comprises 4585 rows, with each row representing a separate field for each season. Table 1 presents a detailed summary of key parameters influencing maize production annually. Each row in this table represents a distinct year, providing insights into the average values of various environmental, soil, and crop growth indicators. Central to this table is the average season duration (ASD) for each year, a critical metric that reflects the time span from planting to harvest. This duration is pivotal in understanding maize growth cycle and how it interacts with environmental factors.

The table also includes average annual precipitation values (prec_T), a primary factor affecting soil moisture and irrigation needs. Solar radiation (srad_T) is reported, indicating the available energy for photosynthesis, essential for crop growth. Soil health indicators such as cation exchange capacity (CEC) and soil organic matter (SOM) are included, representing the soil’s nutrient-holding capacity and overall fertility, respectively. Furthermore, the table outlines the average levels of key nutrients – nitrogen, phosphorus, and potassium. These are crucial for maize growth and development. Maize yield (YIELD) data provides a direct measure of agricultural productivity and farming practices effectiveness each year.

Moreover, the table provides an overview of climatic conditions, including the average maximum and minimum temperatures for each year. These temperatures are vital for understanding crop growth rates and environmental stress factors. Vapor pressure (vp_T) is also included, offering insights into humidity conditions which can impact pest activity and disease prevalence in maize crops. Weather data collected from DAYMET was arranged into thirty ten-day periods covering the period from two months before planting to eight months after planting. Therefore, this table provides a comprehensive understanding of how environmental, soil, and climatic factors, along with crop management

practices, impact maize yields and growth cycles over time.

Typical maize yield values ranged from 0.1 to 9.8 Mg ha⁻¹, with an average of 75 % falling below 5.0 Mg ha⁻¹. The majority of observations were centered around a median elevation of 700 m above sea level. However, elevations varied from near sea level to approximately 300 m. Planting dates exhibited a standard deviation of 20 days, with some observations separated by up to five months. Notably, sighting frequencies showed substantial variation across different years, with fewer observations in 2014 and an increase thereafter.

The average season duration was 175 days. A significant proportion of farmers (greater than 25 %) did not utilize fertilizers containing phosphate, while a majority (greater than 50 %) did not employ fertilizers containing potassium. On average, farmers applied 110 kg of nitrogen ha⁻¹, although some used up to three times that amount. The absence of nitrogen rates was often due to farmers not reporting their applications or incorporating nitrogen from alternative sources such as manure. Conventional tillage was the primary method used, followed by no-till practices. The dataset encompassed approximately 250 different cultivars, many with low frequencies. The number of observations per cultivar varied significantly across years due to high hybrid turnover. A clear regional distribution was observed, with the hybrid system prevailing in the middle west of the state. In contrast, small-scale, self-sufficient farmers were predominant in northern and eastern municipalities. Maize yields in the Landrace system generally ranged between 1.6 and 3.0 Mg ha⁻¹. In contrast, the Hybrid system exhibited yields between 4.2 and 5.3 Mg ha⁻¹.

2.2. Data visualization and correlation

In our study, we analyzed the correlation between various factors as elevation, slope, clay content, planting date, CEC, SOM, N, P, K, pH, Prcp_T, Srad_T, Tmax_T, Tmin_T, Vp_T, and Yield including qualitative variables like cultivar, tillage practices, and cultivation system type. To facilitate this analysis, qualitative variables were first transformed into quantitative data through encoding, assigning unique numerical identifiers to each category. This conversion enabled the inclusion of these factors in a correlation analysis. Using Pearson’s correlation coefficient, we assessed the linear relationship between these now-numerical qualitative variables and maize yield. It is imperative to note that this method measures linear relationships and does not imply causation. The conversion of qualitative to quantitative data for this analysis is a recognized technique in data science, providing comprehensive insights into the impact of various management practices on crop yield.

In Fig. 3, correlation heat maps generated using the Seaborn library show the statistical significance of variables based on their correlation coefficient values. A coefficient of -1 indicates a strong negative association, +1 signifies a strong positive association, and zero implies no significant impact on output conditions. These heat maps visualize interdependencies among factors, providing an overview of the considered output based on input variables. Pearson’s parametric correlation was used to calculate the displayed correlations in Fig. 3.

The yield has shown to have positive correlation with planting, cultivar, Vp_T, Tmax_T, Tmin_T, N, P, and K [40], while having a

Table 1
Annual Averages of Key Agricultural Parameters and Maize Yield (Year wise Summary).

Year	C	N (kg/ha)	P (kg/ha)	K (kg/ha)	CEC (cmolc/dm ³)	SOM (%)	PH	prec_T (mm/day)	srad_T (MJ/m ² /day)	TR (°C)	VP (Pa)	YIELD (Mg/hac)	ASD (days)
2012	123.06	134.81	26.99	6.36	23.58	1.70	6.87	138.21	525.61	7.8–39.2	396–3128	4.50	167
2013	146.81	100.78	26.96	10.43	22.62	1.68	6.91	149.74	519.36	3.7–35.8	328–2740	3.63	163
2014	93.25	104.01	23.99	5.18	23.36	1.67	6.66	115.49	546.27	4.4–35.8	364–2840	2.75	156
2015	113.25	146.39	44.40	31.25	22.40	1.47	6.89	104.0	526.67	8.2–35	476–2656	3.75	179
2016	84.57	99.15	16.15	5.55	22.88	1.62	6.62	110.64	543.68	3.8–36	320–3032	3.13	159
2017	104.76	101.04	20.69	8.21	22.72	1.55	6.89	133.09	525.80	–2.2–36.6	252–3080	3.82	164
2018	106.85	99.59	21.71	7.43	22.41	1.46	6.84	99.01	533.50	4–36.3	292–6452	3.35	165

C: Cultivar, Temp_R: Temperature Range, VP_R: Vapor Pressure Range, and ASD: Average Season Duration.

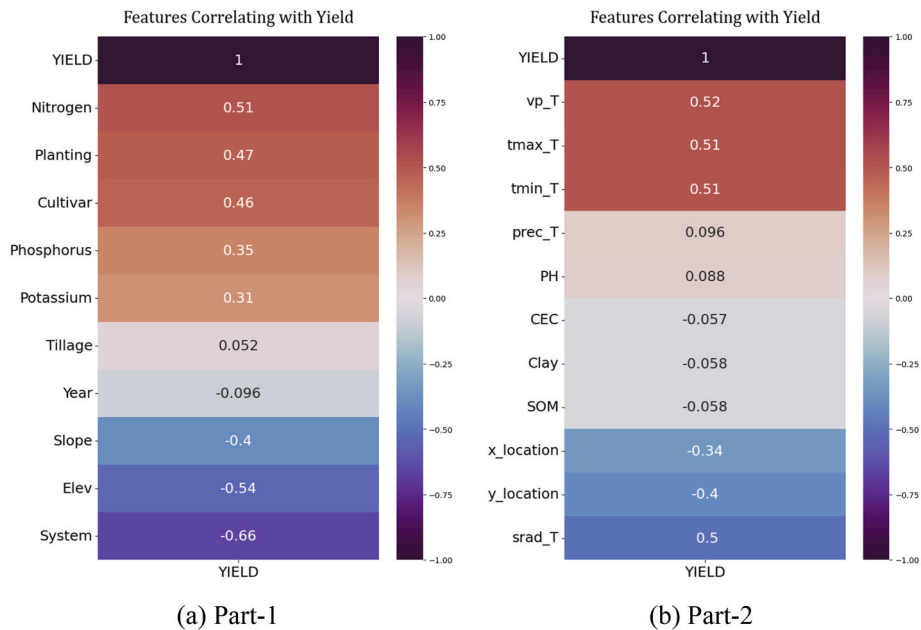


Fig. 3. Correlation charts for output variable (Maize Yield) with other input variables (field management practices, soil parameters, genotype, and ambient weather conditions).

negative correlation with elevation, slope, and system [41]. There is a greater impact on maize yield in the experiment area related to system, elevation, N, Vp_T, temperature, and Srad_T than with tillage, soil quality parameters, and Prcp_T [41,42]. Among the parameters that affect maize yield, system, and elevation have the strongest negative impact, while others have the strongest positive impact [43]. Moreover, the significance of the calculated Pearson correlations was evaluated using a T-test with a significance level of $\alpha = 0.05$. The results indicated that all the Pearson correlations in the study were statistically significant, suggesting that the observed relationships between variables were unlikely to occur by chance. Interestingly, even correlations with values less than 0.3 were statistically significant. This highlights that although these correlations may be weak, they still provide valuable predictive power, as they are part of a larger set of factors that collectively determine output. These findings emphasize the importance of considering multiple variables when analyzing the relationship between predictors and the outcome variable, as even seemingly weak associations can contribute meaningfully to the overall predictive performance of the model.

Furthermore, ANOVA (Analysis of Variance) has also been performed to determine whether there are any statistically significant differences between maize yield means between different years. In interpreting an ANOVA test result, the F-statistic and the p-value are important components. F-statistic is the calculated statistic for the ANOVA test. A higher value generally indicates a more significant difference between the group means. The p-value indicates the probability of obtaining a result at least as extreme as the one observed, assuming that the null hypothesis (which typically states that there are no differences between the groups) is true. If the p-value is extremely low (significantly lower than the common alpha level of 0.05), the null hypothesis can be rejected. This suggests that maize yields differ significantly across the years.

Our analysis of variance (ANOVA) revealed significant differences in maize yields across different years (F-statistic = 47.06, p-value < 0.001). The annual trends in maize yield, as averaged for each year can be visualized in Fig. 4. This result suggests that year-to-year variations significantly affect maize yields, indicating the potential influence of varying environmental conditions, agronomic practices, and other year-dependent factors on crop performance. These findings underscore the

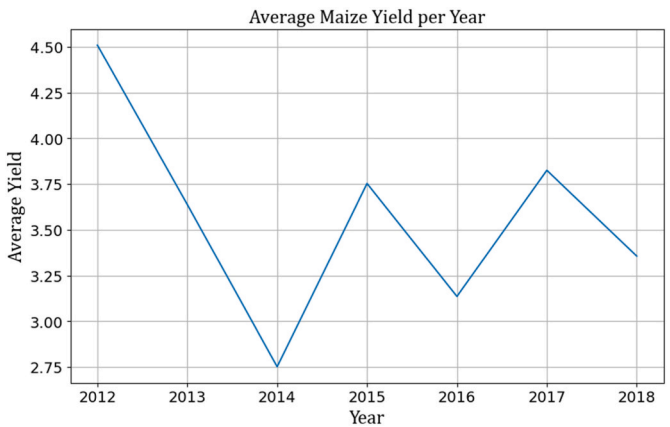


Fig. 4. Annual trends in maize yield.

importance of considering inter-annual variability in agricultural research and planning.

2.3. Exploratory Data Analysis and feature importance evaluation for machine learning model training

For a robust analysis of the dataset and to identify key predictors that affect maize yield and train our model, we utilized Exploratory Data Analysis (EDA) and Feature Importance Evaluation (FEI). We implemented a Random Forest Regressor for feature importance analysis. Random Forest, as an ensemble of decision trees, captures both linear and nonlinear relationships. It does not require any specific form of relationship between predictors and the target variable.

2.3.1. Exploratory Data Analysis (EDA)

Initial EDA was conducted to examine the basic statistical properties and distributions of the dataset. To understand central tendencies and dispersions, summary statistics were calculated for all numerical variables, including means, medians, standard deviations, and interquartile ranges. To ensure data integrity, the types of data in each column and the presence of missing values were also assessed.

Using Seaborn's pairplot function, pairwise plots were generated to visualize relationships and distributions among variables. To identify trends, outliers, and possible correlations between variables, it was imperative to perform this step. Additionally, a correlation matrix, visualized in the form of a heatmap, was used to examine the relationships between the variables. The results provided insights into potential collinearity and the relationships between variables by identifying highly correlated features.

2.3.2. Feature importance analysis with random forest

Following the EDA, we conducted a feature importance analysis using a Random Forest Regressor. The dataset was split into training and testing sets, ensuring a 20 % hold-out for model validation. The Random Forest model, comprising 100 estimators, was trained on the training set with 'YIELD' as the target variable.

Using the feature_importances_ attribute, the mean prediction yield of maize was evaluated post-training. The importance scores were sorted in descending order and visualized through a bar chart in Fig. 5. This provided a clear understanding of which variables had the most significant influence on yield, aiding in refinement of our predictive models.

This combination of EDA and Random Forest-based feature importance evaluation formed the cornerstone of our data analysis approach, ensuring a data-driven and transparent methodology in our study.

2.4. Model selection

To achieve a high level of prediction accuracy, it is crucial to fine-tune the hyperparameters of machine learning models and select the most appropriate models based on their optimized hyperparameter values. Hyperparameters are configuration settings that control the behavior of models, such as learning rates, regularization parameters, or tree depths. By systematically exploring different combinations of hyperparameter values, we can identify the settings that lead to the most accurate predictive performance [44].

An effective approach to evaluating trained models' prediction performance is through cross-validation, a widely employed technique in machine learning. Cross-validation involves dividing the training set into training and validation subsets. The training subset is used to train the models, while the validation subset is used to assess their predictive performance [45].

One popular variant of cross-validation [46] is random cross-validation, where the training set is randomly divided into ten equally sized subsets. The models are then trained and evaluated ten times, each using a different subset as the validation set and the remaining subsets as the training set. This process provides a robust

assessment of the model's performance. It ensures that all data points are used for both training and validation across different iterations. By employing random cross-validation, we can systematically adjust machine learning models' hyperparameters, iteratively refining their settings to improve predictive accuracy. This iterative process helps us find the optimal hyperparameter values that result in the most effective performing models for a given dataset and prediction task (cf. Fig. 6).

2.4.1. Grid search

Grid search [47] is a widely used method for tuning hyperparameters by systematically exploring every possible parameter set in the search space. In a grid search, hyperparameter grids represent multidimensional spaces, with each dimension corresponding to a parameter. These grids create a comprehensive search space that covers all possible hyperparameter combinations. For example, consider two hyperparameters, C and α , with respective values of $C = 100, 150, 200$ and $\alpha = 0.001, 0.01, 0.1$. In the search space, each unique combination of C and α represents a point. For instance, $(C = 150, \alpha = 0.01)$ would be a viable point in the search space. During a grid search, all possible parameter values are considered. A model is created for each combination (point) in the search space. These models are then evaluated based on their performance metrics, such as accuracy or error rates. The goal is to identify the hyperparameter combinations that yield the highest predictive accuracy.

By systematically assessing the performance of each model across different hyperparameter combinations, grid search helps us identify the optimal set of hyperparameters for a given machine learning algorithm. The combinations that result in the highest accuracy or the highest performance are selected as the final results of the grid search. Thus, by considering all possible combinations of hyperparameter values, grid search explores the entire search space. It creates models for each combination and evaluates their performance. This allows us to identify the hyperparameter settings that lead to the most accurate and effective models.

2.4.2. Random search

Similar to grid search, random search [47] is a hyperparameter optimization method that explores the search space. However, instead of exhaustively examining all values, random search randomly selects samples from the upper and lower boundaries of the search space. This approach makes random search more autonomous, parallelizable, and efficient in resource allocation, making it a powerful tool for hyperparameter tuning. In certain cases, random search can outperform grid search, particularly when only a few hyperparameters significantly impact the machine learning algorithm's performance. When there are relatively fewer influential hyperparameters, random search's ability to explore diverse regions of the search space can lead to faster identification of optimal hyperparameter combinations. By leveraging random search's flexibility and adaptability, researchers and practitioners can efficiently navigate the hyperparameter space and discover effective configurations for machine learning models.

2.5. Predictive models

As part of our research, we conducted a thorough evaluation of multiple ML regression models, employing diverse evaluation metrics, with the aim of identifying the optimal performing model. This model would then be used to predict maize crop yields. To ensure accurate predictions and maintain consistent high performance, it is crucial to demonstrate variation in predictions across different models. To achieve this goal, we selected and trained several regression models, including decision tree, Extreme Gradient Boosting (XGBoost), Gradient Boosting, Light Gradient Boosting machine, and random forest. Each of these models offers unique characteristics and strengths that contribute to their effectiveness in predicting crop yield. By carefully assessing and comparing the performance of these models using appropriate

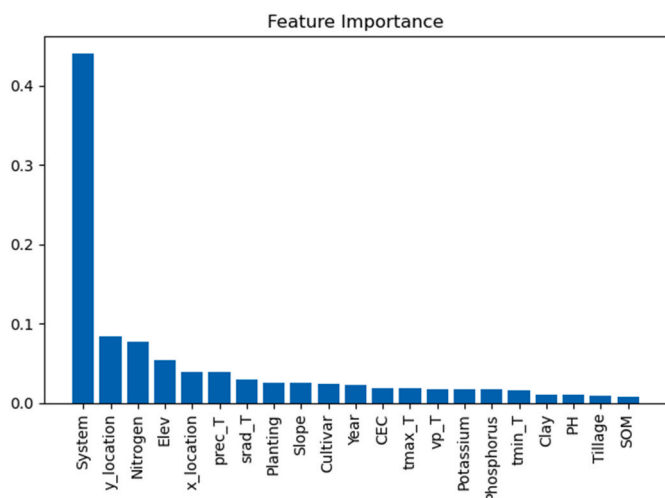


Fig. 5. Relative importance of features in predicting maize yield.

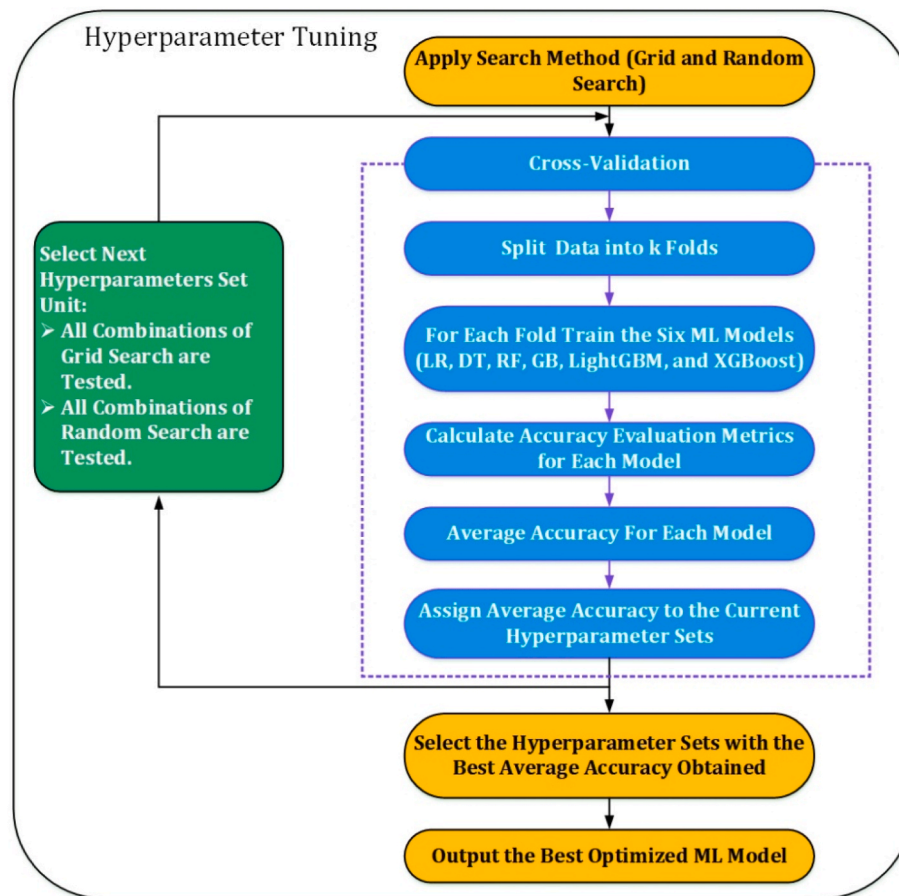


Fig. 6. Comparative analysis of hyperparameter optimization techniques- Random search and Grid search for fine tuning ML models.

evaluation metrics, we can determine which model demonstrates the highest level of accuracy and reliability. This is for the specific problem at hand. This meticulous process enables us to identify the most accurate model that reliably predicts maize crop yield.

2.5.1. Decision tree

A decision tree [48] regression approach is an interpretable and versatile approach to regression problems. The algorithm constructs a tree-like structure based on features and makes predictions by traversing it. It handles both linear and non-linear relationships, accommodates different data types, and is robust against outliers. Overfitting can, however, occur. The performance and generalization of the model can be enhanced by ensemble methods such as gradient boosting regression and random forest.

2.5.2. Gradient boosting

Gradient boosting regression creates a robust predictive model by combining weak regression models, such as decision trees. It effectively captures both linear and nonlinear relationships by iteratively learning from previous models' mistakes [49,50]. With this approach, accurate predictions are made and optimization techniques are utilized to ensure efficient computations. As a result, gradient boosting regression provides reliable results. It is a powerful and flexible method to deal with regression problems.

2.5.3. Random forest

Bagging is an ensemble model based on trees, and random forest is a variant of bagging that builds upon this concept. Bagging reduces prediction variability by averaging predictions obtained from randomly sampled data with replacement. Random forest further enhances

bagging by incorporating feature randomization. In a random forest, the process involves randomly selecting a subset of features and constructing a tree based on these chosen characteristics [51]. This process is repeated multiple times, creating a collection of trees. Ultimately, the predictions from all the trees are averaged to produce the final result. By combining predictions from diverse trees, random forests effectively address both the bias and variance components of error. This comprehensive approach leads to improved performance and generalization in predictive modeling tasks.

2.5.4. LightGBM and XGBoost

These are both gradient-boosting tree-based ensemble methods. These approaches utilize sequential ensembles to combine weak predictive tree models and iteratively learn from their errors during the prediction process. XGBoost [52], introduced in 2016, incorporates additional features, while LightGBM [53], released by Microsoft in 2017, offers enhanced performance and faster computation times [45]. Both LightGBM and XGBoost excel at managing sparse data, a crucial capability for efficient processing of datasets with many zero values. Moreover, they leverage approximation techniques to optimize performance, achieving superior results in accuracy and efficiency. Overall, LightGBM and XGBoost represent advancements in gradient boosting methods. They provide powerful tools for predictive modeling tasks, particularly when working with sparse data and demanding computational requirements.

2.6. Optimized model

In our study, we employed a comprehensive approach of hyperparameter tuning and model selection for various regression models,

including decision tree, Extreme Gradient Boosting (XGBoost), Gradient Boosting, Light Gradient Boosting machine, and random forest. The goal was to identify the model that exhibited the highest prediction performance. To accomplish this, we utilized grid search and random search techniques coupled with cross-validation [54,55]. These approaches enabled us to systematically explore the hyperparameter space and evaluate different models for each regression algorithm. Through this process, we evaluated multiple scoring metrics, such as R-squared (R^2), mean squared error (MSE), and mean absolute error (MAE), to assess the prediction accuracy and robustness of the models.

After conducting the hyperparameter tuning and model selection process, the evaluation metrics clearly indicated that Extreme Gradient Boosting (XGBoost) consistently outperformed the other regression models. Grid search-based XGBoost demonstrated superior performance in terms of both accuracy and computational efficiency, making it the most suitable choice for our prediction task. By selecting Grid search-based XGBoost as the final model, we ensured that our regression model could effectively capture the underlying patterns in the data and provide highly accurate predictions for the desired quantitative response. Overall, our approach of hyperparameter tuning and model selection, which involved exploring multiple regression algorithms, demonstrated that Grid search-based XGBoost yielded the highest prediction performance. This finding reinforces the effectiveness and superiority of XGBoost in regression tasks and supports its selection as the optimal model for the given problem.

3. Results and discussion

In this section, we delve into the impact of various crop management practices on maize crop yield and present the results of optimized machine learning-based predictive modeling. We explore how different agricultural techniques and strategies affect maize crop yield. Additionally, we showcase the outcomes of our optimized machine learning models, which have been fine-tuned to provide accurate maize yield predictions.

3.1. Data analysis results

3.1.1. Impact of pH, precipitation, temperature, solar radiations, and vapor pressure on maize yield

This analysis investigates the relationships between various parameters and their impact on maize yield. To achieve this, scatter plots were generated by grouping the five parameters, namely, pH, Prcp_T, T_{max}_T, T_{min}_T, and Vp_T, and calculated the mean of maize yield (YIELD) for each group. Scatter plots offer a valuable method for examining the association between two variables and identifying potential patterns or trends within the dataset. By examining scatter plots, we can gain insights into how different factors may influence maize crop yield. This visual exploration provides a foundation for understanding the interdependencies between parameters and their effects on agricultural yields.

In the scatter plots, the color hue represents the maize yield. Darker colors indicate higher yields and lighter variations representing lower yields. By examining these scatter plots, we can gain insights into the impact of different factors on maize yield. The Seaborn library in Python was utilized to create the scatter plots. Each scatter plot depicts the relationship between two parameters, with the x-axis and y-axis corresponding to the selected parameters. The visual representation allows for a comprehensive understanding of the interplay between these parameters and their impact on maize yield.

The scatter plots presented in Fig. 7 depict the relationship between the soil pH level, precipitation, solar radiation, and maximum and minimum temperature in relation to maize yield (YIELD). Upon analysis of these scatter plots, it is evident that the pH values within the range of 5–8 do not exhibit a significant impact on the influence of these parameters on the crop yield. However, when examining the relationship between precipitation and pH, it becomes apparent that the maximum yield is observed within the precipitation range of 5–15 mm day⁻¹. Similarly, the scatter plot illustrating the association between soil pH and solar radiation indicates that solar radiation values ranging from 16 to 20 MJ m⁻² day⁻¹ correspond to the highest yield. Upon further examination of the relationship between soil pH and maximum and minimum temperatures, a scattered pattern is observed among the data

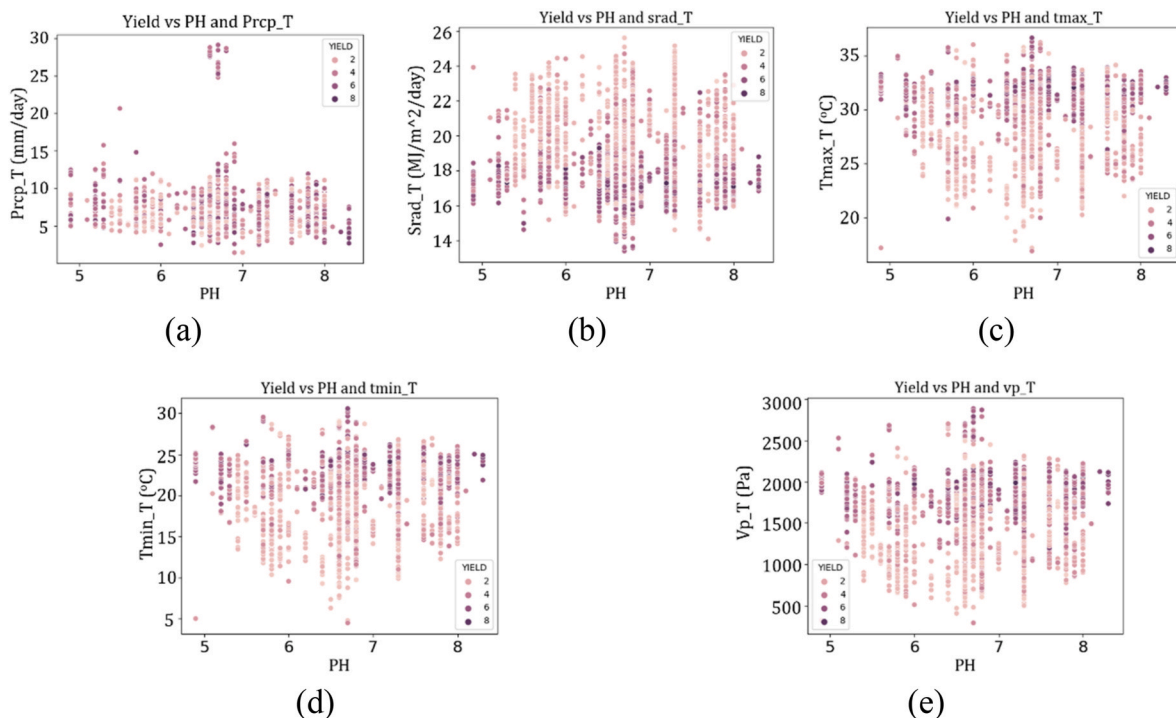


Fig. 7. Scatter plots illustrating the relationship between pH and Yield with Prcp_T, T_{max}_T, T_{min}_T, and Vp_T parameters.

points. This suggests that there is no clear linear trend between pH and temperature, implying that pH level does not strongly influence the effect of temperature and vapor pressure on crop yield. However, upon closer inspection, it is noticeable that temperature and vapor pressure within the ranges of 25–35 °C and 1500–2000 Pa, respectively, exhibit a more favorable trend for achieving higher crop yield.

Ensuring the correct soil pH is of the utmost importance as acidic or alkaline soils can constrain nutrient availability, impacting maize growth and yield. Furthermore, sufficient and evenly distributed rainfall cannot be overstated. Drought conditions and excessive moisture can both detrimentally affect maize yields. To address these challenges, the implementation of intelligent water management strategies and the cultivation of drought-resistant maize varieties can effectively mitigate these adverse effects.

Solar radiation plays a pivotal role in driving photosynthesis and fostering maize growth. Nonetheless, excessive radiation combined with elevated temperatures may induce heat stress. Therefore, achieving a harmonious balance between sunlight and temperature becomes essential for optimal yields.

Temperature has a profound influence on maize growth stages. Elevated temperatures during critical stages, such as flowering and grain-filling, have the potential to significantly reduce yields. Consequently, the selection of maize varieties adapted to prevailing temperature conditions, as well as the utilization of climate-appropriate planting techniques, are crucial to maximizing maize yields.

The scatter plots presented in Fig. 8 depict the relationship between the solar radiations, temperature, and vapor pressure in relation to maize yield. Upon analysis, it is evident from the scatter plots that the yield data points are predominantly distributed across a range of 5–20 mm day⁻¹ for precipitation. However, upon closer examination of the Srad_T vs. Prcp_T plot, it becomes clear that the data points associated with high yields are concentrated within the narrower range of 16–20 MJ m⁻² day⁻¹ for solar radiation. This indicates that solar radiation levels within this specific range have a more pronounced positive impact

on crop yield. Similarly, when analyzing the relationship between precipitation (Prcp_T) and yield, it is observed that the data points are spread across various precipitation levels. However, a closer look reveals that the points associated with high yields are primarily concentrated within the narrow range of 5–15 mm day⁻¹ for precipitation. This highlights the significance of precipitation levels within this specific range for achieving optimal maize yield.

Furthermore, the scatter plot for precipitation and temperature shows a scattered distribution of data points. The data points cover a range of 22–35 °C for maximum temperature (Tmax_T), but the high yield points are concentrated in the range of 30–35 °C for Tmax_T and 20–25 °C for minimum temperature (Tmin_T), along with a precipitation range of 5–15 mm day⁻¹. This suggests that a combination of high maximum temperature, moderate minimum temperature, and specific precipitation levels contribute to increased crop yields. Similarly, the scatter plot for precipitation and vapor pressure exhibits a scattered pattern. The data points cover a range of 500–2500 Pa for vapor pressure (Vp_T), but the high yield points are concentrated in the range of 1500–2000 Pa for vapor pressure and the precipitation range of 5–15 mm day⁻¹. This indicates that specific levels of vapor pressure, along with moderate precipitation, play a significant role in achieving higher crop yields.

The scatter plot in Fig. 9 showcases the correlation between solar radiation (Srad_T) with Tmax_T, Tmin_T, and Vp_T in relation to the maize yield. When examining the scatter plot between solar radiation (Srad_T) and maximum temperature (Tmax_T), a clear upward trend is observed among the data points. This indicates that higher solar radiation combined with higher maximum temperatures positively influences maize yield. This finding suggests that solar radiation plays a crucial role in maximizing crop productivity when accompanied by elevated maximum temperatures. Moving on to the relationship between solar radiation (Srad_T) and minimum temperature (Tmin_T), a linear distribution of data points is observed. However, the lack of a clear linear trend suggests that solar radiation may not significantly impact yield in

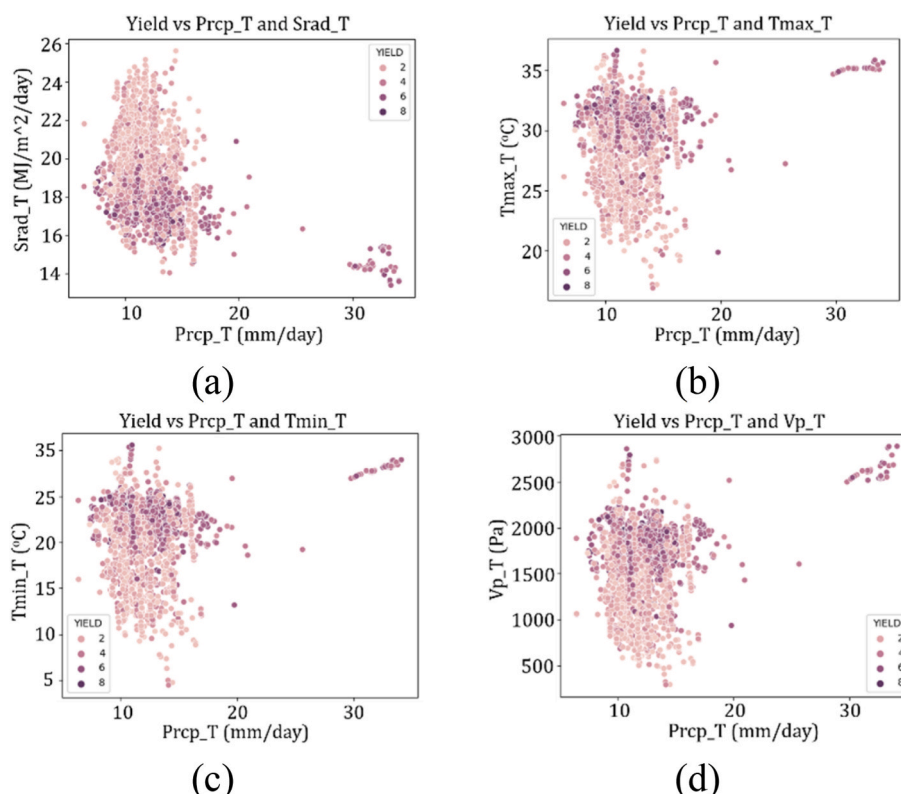


Fig. 8. Scatter plots illustrating the relationship between Prec_T and Yield with Srad_T, Tmax_T, Tmin_T, and Vp_T parameters.

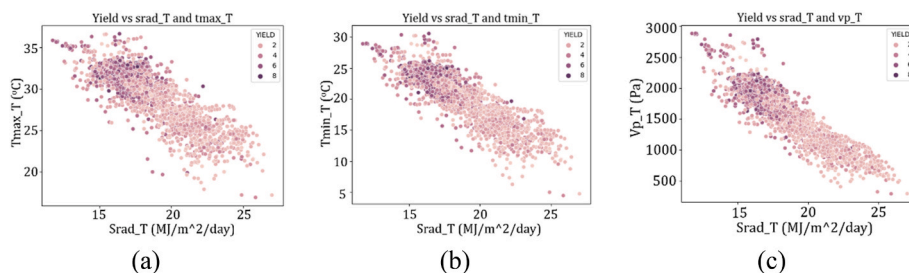


Fig. 9. Scatter plots illustrating the relationship between Srad_T and Yield with Tmax_T, Tmin_T, and Vp_T parameters.

conjunction with minimum temperature.

Other factors, such as precipitation and soil moisture, may have a more dominant influence on the relationship between minimum temperature and maize yield. In the scatter plot between solar radiation (Srad_T) and vapor pressure (Vp_T), the data points appear scattered, indicating the absence of a strong linear relationship between solar radiation and vapor pressure. This suggests that solar radiation may not have a significant direct impact on the relationship between vapor pressure and maize yield. Other factors, such as soil moisture availability and plant transpiration, might play a more critical role in determining yield response to vapor pressure.

Furthermore, Fig. 10 illustrates how temperature and vapor pressure influence maize yield. Examining the correlation between maximum temperature (Tmax_T) and minimum temperature (Tmin_T) with respect to yield, we observe a linear distribution of data points, indicating a strong positive linear trend. The difference between maximum and minimum temperatures appears to have a significant influence on crop yield. Similarly, the scatter plot depicting the relationship between maximum temperature (Tmax_T) and vapor pressure (Vp_T) shows a linear distribution of data points. The positive linear trend suggests that vapor pressure's impact on yield is significantly influenced by maximum temperature. Additionally, analyzing the association between minimum temperature (Tmin_T) and vapor pressure (Vp_T) reveals another linear distribution of data points. This suggests that the relationship between minimum temperature and vapor pressure strongly affects maize yield. The range of values for these parameters, where high yield values are observed, aligns with the ranges discussed in the previous scatter plots.

Therefore, the above analysis underscores the complex interplay of environmental factors such as soil pH, precipitation, solar radiation, and temperature in influencing maize yield. Effective yield optimization requires managing these interconnected elements, including maintaining suitable soil pH for nutrient absorption, implementing precise water management, and adapting planting to local climatic conditions, particularly solar radiation, and temperature. While soil pH doesn't significantly directly affect yield, optimal levels of precipitation and solar radiation, especially when combined with higher maximum temperatures, are vital for boosting productivity. However, the impact of solar radiation on yield is less significant in conjunction with minimum temperature and vapor pressure, suggesting that soil moisture and other factors may be more impactful here. Crucially, temperature variations,

especially the difference between maximum and minimum temperatures, play a pivotal role in yield maximization. This necessitates the selection of maize varieties tailored to regional climates and the integration of advanced agronomic practices, including precision agriculture and the development of resilient crop varieties.

3.1.2. Impact of system, tillage, cultivar (genotype), and planting time on maize yield

The impact of various farm management practices on maize yield is examined in this subsection, including system, tillage practices, cultivar (genotype), and planting dates. Understanding how these factors influence maize productivity is essential for optimizing agricultural practices and maximizing crop yields. By analyzing their effects, we aim to gain insights into the key determinants of maize yield and provide valuable information for farmers and researchers to enhance maize production strategies.

The Grouped Bar Chart presented in Fig. 11 offers valuable insights into two distinct maize farming systems: the Landrace-based and the Hybrid system. The Landrace-based system characterizes small-scale, low-input self-consumption farmers, while the Hybrid system represents medium-scale, medium-input mechanized semi-commercial farmers engaged in local market sales. This visualization provides a

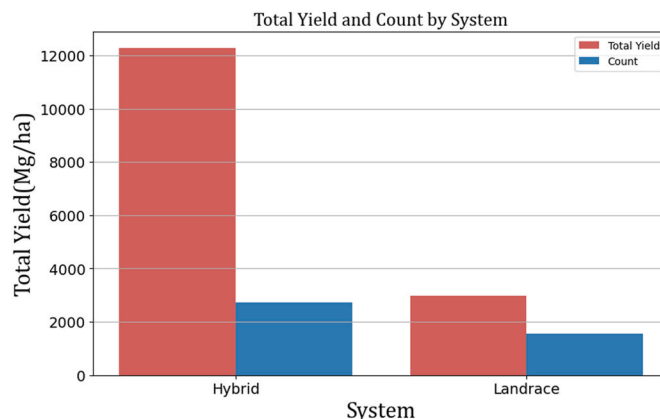


Fig. 11. Grouped bar chart comparing total maize yield by farming systems.

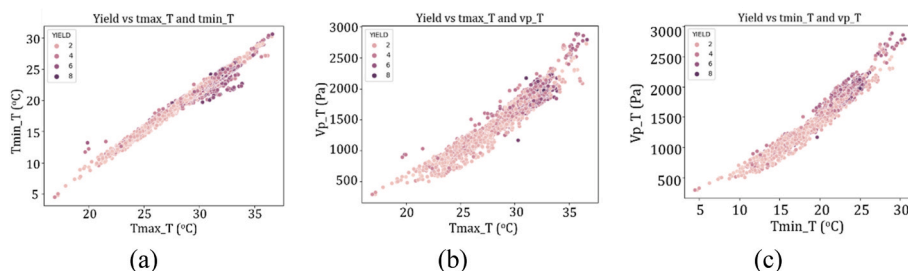


Fig. 10. Scatter plots illustrating the relationship between Tmax_T and Yield with Tmin_T and Vp_T and Tmin_T and Yield with Vp_T parameters.

comprehensive overview of total yield and occurrence count for each system.

On the x-axis, the chart labels indicate the systems under comparison, distinguishing between the Landrace-based and Hybrid systems. Meanwhile, the y-axis labels signify the corresponding values for total yield and count. Notably, the Grouped Bar Chart vividly illustrates a significant disparity in total yield between these two systems. Specifically, the hybrid system demonstrates exceptional performance, producing approximately four times the yield of the Landrace-based system.

These findings underscore the positive impact of implementing mechanized techniques and medium-input practices, with a focus on commercial production, on maize yield. Our subsequent recalculations, considering yield per unit of cultivated land and accounting for occurrence frequencies, reaffirm these substantial differences between the two systems. This analysis further emphasizes the advantages associated with the Hybrid system in terms of maize yield and suggests potential avenues for enhancing productivity within different farming contexts.

As depicted in Fig. 12, various tillage practices influence maize yield. The Grouped Bar Chart offers a comprehensive overview of both total yield and occurrence counts for each specific tillage practice. Our analysis focuses on three primary tillage methods: conventional tillage, no tillage, and reduced tillage. The chart effectively conveys the marked variations in total yield among these tillage practices. On the x-axis, the chart labels distinguish the specific tillage methods under comparison, while the y-axis labels represent the corresponding values for both total yield and count.

When examining the occurrence frequencies of different tillage practices and recalculating the total yield per unit of cultivated land, we find significant differences in yield performance. The analysis of tillage data in our study presents intriguing insights into the impact of different tillage practices on maize yield. Our findings indicate a marginal yet noteworthy difference in yield between conventional, no-tillage, and reduced-tillage practices. Specifically, conventional tillage practices resulted in an average yield of 3.455 mg per hectare. No-tillage and reduced-tillage practices yielded slightly higher amounts of 3.662 mg and 3.685 mg per hectare, respectively. This data suggests a potential advantage of no-tillage and reduced tillage methods over conventional tillage in terms of crop yield. The higher yields associated with no-tillage and reduced tillage could be attributed to factors such as better soil moisture retention, decreased soil erosion, and improved soil structure and health, which are known benefits of these practices.

However, it is crucial to consider the disparity in the number of instances for each tillage method observed in our dataset as presented in Fig. 12. Conventional tillage was the most frequently used method with 2092 instances, followed by no-tillage with 1553 instances, and reduced tillage with only 769 instances. This variation in use frequency could be influenced by several factors, including regional farming practices, availability of equipment, and farmers' knowledge and perception of

tillage practices. The higher number of instances for conventional tillage might reflect its long-standing prevalence in agricultural practices, while the lower instances for reduced tillage could indicate a relatively recent adoption or limited access to resources needed for this practice. The differing number of instances might also impact our findings' generalizability. While reduced tillage shows the highest yield, the smaller sample size compared to conventional tillage could limit confidence in extrapolating this result to a broader context. Similarly, although no-tillage shows a higher yield than conventional tillage, the difference in sample sizes must be considered when interpreting these results. Thus, this analysis underscores the impact of tillage practices on maize yield but also provides valuable insights into optimizing agricultural strategies for enhanced productivity.

During the seven experimental seasons, 250 cultivars were evaluated, each with varying occurrence frequencies and different management practices, resulting in diverse maize yields. However, conducting analysis and visualization based on unbalanced data for each cultivar category, without considering the number of occurrences for each type, can potentially lead to misleading conclusions. This is particularly true when certain cultivars exhibit significantly higher prevalence than others. To ensure accurate and reliable conclusions, it is essential to consider the relative frequencies of each cultivar type when analyzing and visualizing the data.

To address imbalanced data and ensure a fair representation of cultivar categories, an oversampling technique was employed. The goal was to balance the dataset by adjusting the count for each cultivar category to a desired count of 420. This was achieved by randomly duplicating existing data within each category to increase representation of underrepresented cultivars. The oversampling process involved determining the number of additional samples needed for each category to reach the desired count. Existing data points were randomly selected and duplicated accordingly. The resulting balanced dataset comprised the original data combined with the oversampled data, ensuring an equal count of 420 for each cultivar category.

Using this balanced dataset, the yield for each cultivar category was recalculated by summing the yield values of the samples within that category. By recalculating the yield values based on the balanced dataset, we ensured that the total yield for each cultivar category accurately reflected the yield contribution of all the samples within that category. This step was crucial in providing a fair assessment of each cultivar's productivity, considering the adjusted counts achieved through oversampling. The recalculated yield values served as the basis for analysis and visualization. This allowed for meaningful comparisons and identification of the top yielding cultivars in the balanced dataset.

It is imperative to note that this oversampling approach, along with the recalculated yield values, mitigates the potential for misleading conclusions based solely on the number of occurrences of each cultivar category. The adjusted counts and recalculated yields ensure a more equitable representation, enabling more accurate comparisons and analyses of cultivar performance.

Fig. 13 represents the total yield of the top 30 maize cultivars in a balanced dataset. Among these cultivars, the top 11 cultivars (DEKALB, SP20 A, REGA, CB37, VS-535, P4066W, MEZCLA, H-526, P3961W, American seeds, and NB7) are the highest yielding maize types. These cultivars produce the highest yields compared to the other cultivars in the dataset. On the other hand, the remaining cultivars in the dataset exhibit comparable yields to each other. This information suggests that while the top 11 cultivars stand out for their high yield performance, the other cultivars can still provide competitive yields within the dataset. Therefore, depending on other factors such as specific traits, adaptability, and availability, these comparable-yield cultivars might also be viable options for maize production in tropical weather.

In the current study, maize planting has been conducted over an extended period, spanning from early March to early August, as depicted in Fig. 14. However, the analysis reveals that optimal crop yield occurs during early June to late July. This timeframe aligns with favorable

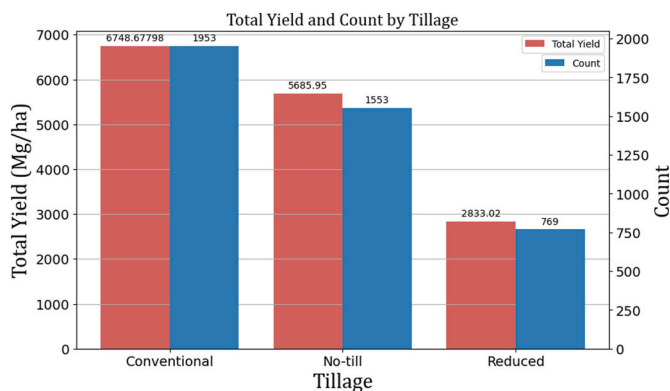


Fig. 12. Grouped bar chart showing maize yield and the occurrence count variation across tillage practices.

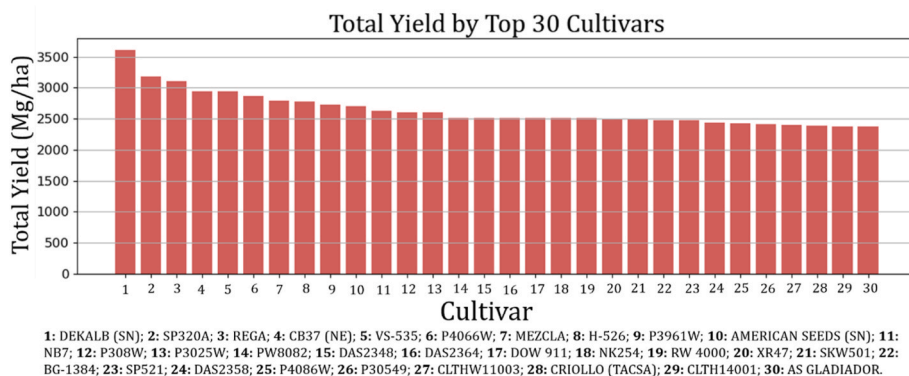


Fig. 13. Total maize yield by Cultivar.

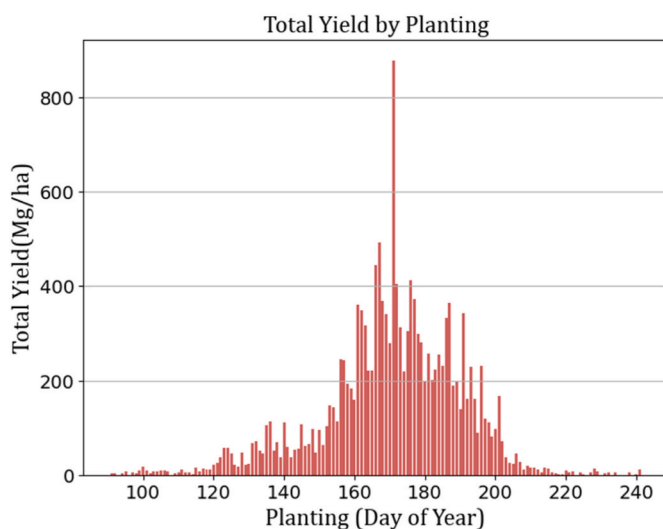


Fig. 14. Total maize yield by Planting (Day of Year).

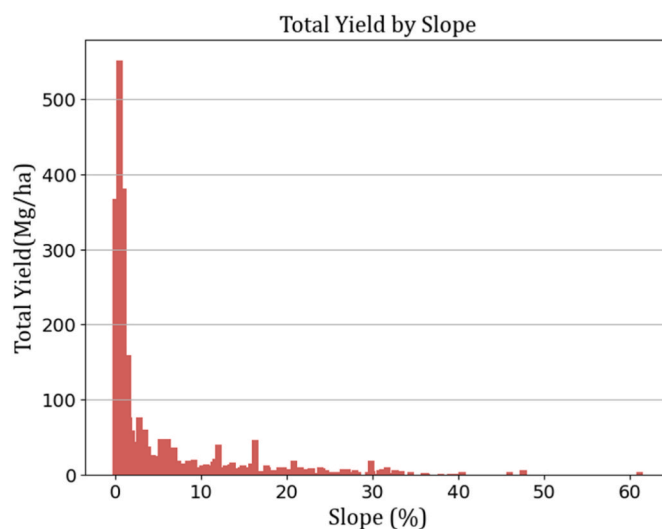


Fig. 15. Total maize yield by slope.

growing conditions for the maize crop, including optimal soil moisture availability and soil temperature for seed germination. During this period, the soil moisture content is typically adequate, facilitating germination and providing the necessary moisture for early plant growth. Additionally, the soil temperature is within the optimum range for seed germination, ensuring the successful emergence and establishment of maize plants. Pearson's correlation coefficient of $r = -0.47$ confirms the relationship between planting date and maize productivity. By focusing on planting within this preferred timeframe, farmers can benefit from the favorable combination of soil moisture and temperature conditions. This will maximize the chances of higher yields and overall crop success.

3.1.3. Impact of slope and fertilization on the maize yield

This subsection assesses the impact of slope and fertilizers on maize production. It is crucial to comprehend the influence of these factors on maize production to optimize agricultural practices and enhance crop yields. This analysis provides valuable insights to enhance maize production techniques. The goal is to improve understanding of key factors that significantly affect maize production and implement effective strategies to improve performance.

The analysis conducted in this study provides valuable insights into the relationship between maize cultivation and slope positions in the experimental area, as depicted in Fig. 15. Concentrating maize production in the footslope position leads to higher grain yields than other slope positions. This is due to soil leaching and nutrient loss in upper slope positions, as well as favorable moisture conditions in lower slope

positions. Fig. 15 illustrates that slopes up to approximately 3 % exhibit the highest maize yield. The Pearson's correlation coefficient value of $r = -0.40$ further supports the relationship between slope position and maize productivity. These results emphasize the importance of considering slope position in maize cultivation strategies, as it significantly impacts grain yields.

Scatter plots (cf. Fig. 16) were generated to explore the relationships between fertilizer inputs and maize yield. The objective was to gain insights into how these parameters influence crop yield and identify potential patterns. Strong positive associations were observed between maize yield and fertilizer inputs for nitrogen (N), phosphorus (P), and potassium (K) in the study area's farms, particularly the larger hybrid system farms. Various tillage practices, such as conventional tillage, no-tillage, and reduced tillage, as well as varying fertilizer levels, were compared on maize yield and growth parameters. Within each landscape position, subplots received varying levels of nitrogen treatments to evaluate the effect of nitrogen application on crop performance. The nitrogen treatments encompassed three levels: no nitrogen (0 kg ha^{-1}), moderate nitrogen (167 kg ha^{-1}), and high nitrogen (349 kg ha^{-1}). Similarly, potassium and phosphorus fertilizers were applied at optimal rates to examine their influence on grain yield. The potassium treatments comprised of no potassium application (0 kg ha^{-1}), moderate potassium application (45 kg ha^{-1}), and high potassium application (99.6 kg ha^{-1}). The phosphorus treatments consisted of no phosphorus application (0 kg ha^{-1}), moderate phosphorus application (68 kg ha^{-1}), and high phosphorus application (143 kg ha^{-1}).

Our data analysis results have provided valuable insights into the

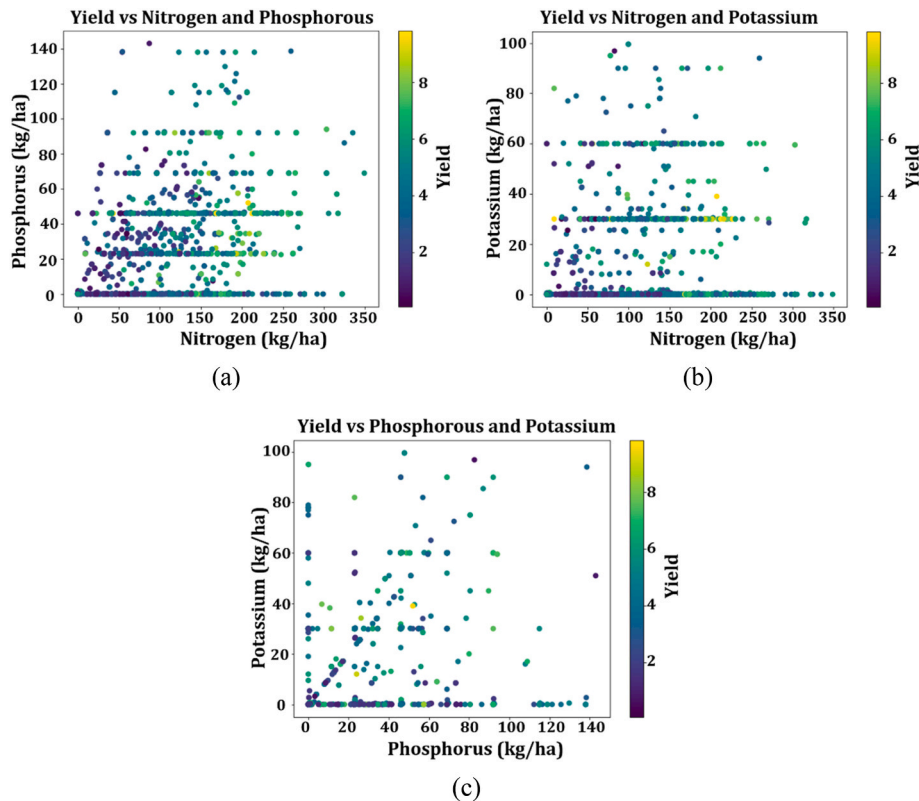


Fig. 16. Scatter plots illustrating the relationship between nitrogen, phosphorus, potassium with maize yield.

impact of nitrogen, phosphorus, and potassium on maize yield. The visual representation and statistical analysis revealed significant effects of fertilizer regimes on grain yield across the sites, as indicated by Pearson's correlation coefficients (r) of 0.51 for nitrogen, 0.35 for potassium, and 0.31 for phosphorus. These coefficients suggest a meaningful relationship between these nutrients and maize yield. One notable finding is the substantial influence of nitrogen application on maize yield. In particular, the high fertilizer treatment, with a focus on nitrogen application, resulted in a remarkable 150 % increase in grain yield compared to the no-nitrogen condition. This observation underscores the critical role of nitrogen in promoting maize growth and productivity. However, it's imperative to acknowledge that low yields were observed in a few instances. This variability could indicate the influence of other factors on crop performance in those cases.

Additionally, our analysis highlighted the importance of phosphorus and potassium. Adequate levels of phosphorus and potassium were associated with improved maize yields. These nutrients play crucial roles in early root development, water and nutrient uptake, and overall plant health. Their positive impact on yield reaffirms their significance in maize cultivation.

Furthermore, the analysis revealed an intriguing insight related to landscape position. The footslope position exhibited an 80 % higher grain yield than the upslope position. This finding underscores the influence of landscape position on crop performance, with topographical factors impacting water distribution, soil characteristics, and ultimately, maize productivity.

Therefore, our data analysis results confirm the substantial impact of nitrogen, phosphorus, and potassium on maize yield. Nitrogen application, in particular, stands out as a key driver of yield enhancement. However, the presence of low yields in specific instances suggests the involvement of additional factors that warrant further investigation. Moreover, the influence of landscape position on maize yield highlights the need for site-specific farming strategies. Overall, these findings provide valuable guidance for optimizing fertilizer applications and

improving crop management practices to maximize maize yields while considering local environmental factors.

These results also underscore the importance of a holistic approach to agricultural management, integrating practices such as tillage systems, choice of cultivars, planting timing, and precise fertilizer applications. A comprehensive understanding of these elements not only advances productivity but also fosters sustainability in farming systems. This integrated perspective is key to unlocking the potential of agricultural practices for higher yield outcomes while maintaining ecological balance.

3.1.4. ANOVA analysis of factors influencing maize yield

The ANOVA analysis was performed to investigate the influence of various environmental and management factors on maize yield. The ANOVA results (Table 2) how the significance of various factors affecting maize yield. Parameters such as Elevation, System, Nitrogen,

Table 2
ANOVA table for maize yield analysis.

Parameter	Sum of Squares	DF	F-value	P-value	
Elevation	51.622072	1	32.506771	1.268202e-08	***
System	582.561646	1	366.843037	1.679702e-78	***
Cultivar	2.788646	1	1.756029	1.851917e-01	**
Tillage	2.202161	1	1.386716	2.390263e-01	*
Planting Time	17.468973	1	11.000331	9.185976e-04	**
Nitrogen	529.832620	1	333.639211	8.055090e-72	***
Phosphorus	31.336784	1	19.732986	9.131203e-06	***
Potassium	20.296682	1	12.780959	3.540527e-04	***
Slope	110.702726	1	69.710261	9.183696e-17	***
Clay	1.214983	1	0.765083	3.817919e-01	
CEC	0.440773	1	0.277558	5.983337e-01	
SOM	2.115989	1	1.332452	2.484342e-01	
PH	1.335421	1	0.840924	3.591846e-01	
prec_T	0.104430	1	0.065760	7.976258e-01	
srad_T	0.028381	1	0.017872	8.936573e-01	
tmax_T	3.470567	1	2.185440	1.393953e-01	*

Phosphorus, Potassium, Slope, and Planting were found to have significant effects, as indicated by their respective p-values and F values. The significance levels are denoted by asterisks: for $p < 0.05$, * for $p < 0.01$, and *** for $p < 0.001$. These findings highlight the importance of these factors in influencing maize yield and underscore the need for precise management of these variables in agricultural practices.

The analysis indicates that elevation (Elev), the farming system used (System), planting strategy (Planting), and the application of nitrogen (Nitrogen), phosphorus (Phosphorus), and potassium (Potassium) fertilizers, as well as the slope of the land (Slope), significantly affect maize yield. These factors all have p-values well below the 0.05 threshold, demonstrating their substantial influence on crop productivity. Conversely, other parameters such as soil characteristics (Clay, CEC, SOM, PH) and climatic variables (prec_T and srad_T) do not exhibit a statistically significant impact on maize yield in this study.

These results underscore the importance of considering key factors such as elevation, farming system, planting strategy, and the balanced application of essential fertilizers in optimizing maize yield. Future research could further explore the interactions between these significant factors and other environmental variables to develop more nuanced and effective crop management practices.

3.2. Predictions

Based on the observed maize yields, we trained 23 maize variables using six machine learning models as summarized in Table 3. These models encompassed linear regression, decision trees, gradient boosting, random forests, and XGBoost. To evaluate the outcomes, we employed a cross-validation approach and summarized the results using different models and data combinations. Considering all the assessment markers, namely R^2 and RMSE, the XGBoost and random forest (RF) models demonstrated the highest prediction accuracy, with an R^2 ranging from 0.83 to 0.86 and a lower RMSE of 0.48.

Comparatively, the remaining models exhibited R^2 values above 0.62; however, their RMSE values exceeded 0.85. This suggests a negligible correlation between these models and the observed yields. Hence, the XGBoost and RF algorithms prove to be more suitable for predicting maize yields in comparison to the other algorithms investigated. Additionally, our analysis revealed that the accuracy of the models varied depending on the combinations of data, even when using the same machine learning method. By utilizing the XGBoost machine learning model, we successfully established a robust model for predicting maize yield in the study region.

The XGBoost and random forest (RF) models, trained in the previous section, were employed to forecast the maize yield. Fig. 17 depicts a scatter plot illustrating the actual and projected yields of the maize management experiment, spanning seven years from 2012 to 2018. Our analysis, with an R^2 value of 0.86, revealed a strong correspondence between the projected and observed yields. These findings indicate the efficacy of machine learning (ML) models in accurately predicting maize production at the field level, with XGBoost demonstrating superior accuracy compared to RF models. However, despite the predicted and measured yields being nearly identical, we encountered certain contradictory data points surrounding the actual yield values.

Table 3

Prediction accuracy of five machine learning regression model for maize yield prediction.

Model	Mean Squared Error	R-Squared
Decision Tree	0.86	0.70
Gradient Boosting	0.87	0.74
LightGBM	0.85	0.74
Random Forest	0.48	0.83
XGBoost	0.48	0.86

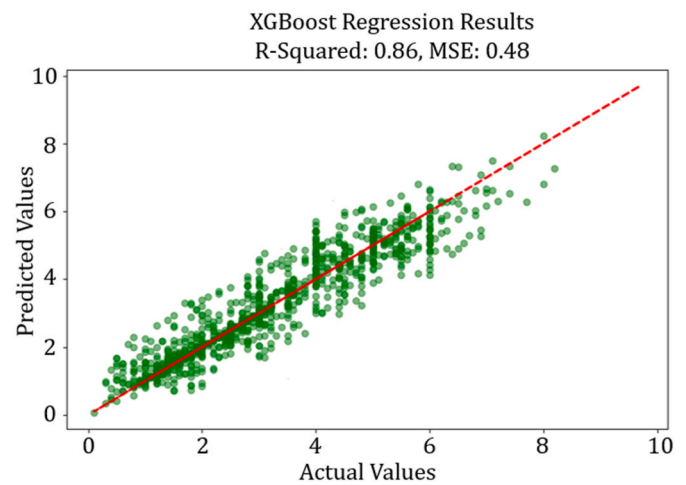


Fig. 17. Scatter plot illustrating the relationship between actual and predicted maize yields using the XGBoost model, depicting crop management systems at the field level during the period from 2012 to 2018.

Fig. 18 reveals that the errors between the predicted and actual values in the XGBoost regression model are primarily concentrated around the zero error value, indicating a good overall fit. On the X-axis, labeled 'Error Values', the plot shows the discrepancy between predicted and actual maize yields, while the Y-axis, marked as 'Error Density', displays the frequency and spread of these errors. Most errors are concentrated around zero, suggesting a good fit, as indicated by the narrow distribution range spanning from -2 to 3. This concentration near zero signifies that the model predictions are generally accurate, with deviations being minimal. Such visualization is crucial for evaluating the model's performance, indicating areas of close alignment between predicted and actual yields, as well as pinpointing where significant discrepancies occur.

3.3. Parameter sensitivity analysis using Explainable Artificial Intelligence (XAI)

XAI (Explainable Artificial Intelligence) provides a range of tools for modeling artificial neural networks, including force plots and summary plots that offer detailed insights into the impact of each parameter [51,

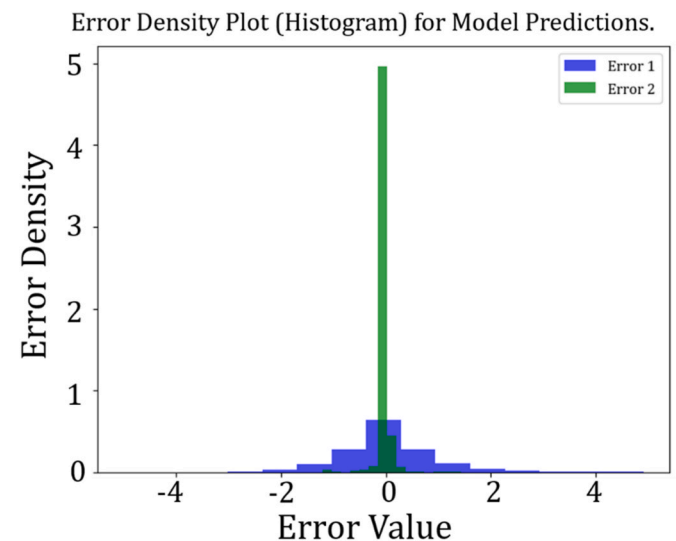


Fig. 18. Error Density plot of XGBoost regression model illustrating concentrated errors around zero with a range of -2 to 3. Indicates areas with high aligned predictions and high deviations.

56]. The force and summary plots were generated using the pertinent functions available in the SHAP library, which were employed in our study. These plots provided us with a comprehensive understanding of the importance of various features in shaping our model's predictions. Specifically, the force plot showcases how each characteristic influences the prediction of a particular observation. In Fig. 19, the force plot demonstrates the influence of each parameter on the observations in the test dataset. The plot uses the test data to indicate the impact of each parameter on every observation. Notably, positive effects are represented by highlighted pink values, while negative effects are denoted by highlighted blue values. This model exhibits resilience in predicting test results by considering input characteristics with diverse positive and negative impacts. Consequently, it unveils distinct variations based on the input characteristics for each observation. Additionally, it is possible to generate force plots for individual observations separately. In Fig. 19

(b–h), the force charts provide a more detailed view of the individual test data points presented in Fig. 19(a).

The force plot for maize yield depicts the variables that have the greatest impact on the model's predictions. Variables that increase the prediction are highlighted in pink, while those that reduce the prediction are shown in blue. The numbers on the black line represent the maize yield in Mg ha^{-1} , based on the training set's base value (3.57), and the corresponding output values from the model's predictions in different individual force plots (Fig. 19(b–h)).

In (Fig. 19(a)), the system parameter is observed as the variable that contributes the most to increasing the model's prediction, indicated by the extended bar associated with this variable. It is followed by variables representing nitrogen application, slope, elevation, and cultivars used. Conversely, precipitation (Prcp_T) is the variable that has the most significant negative impact on the model's prediction, followed by

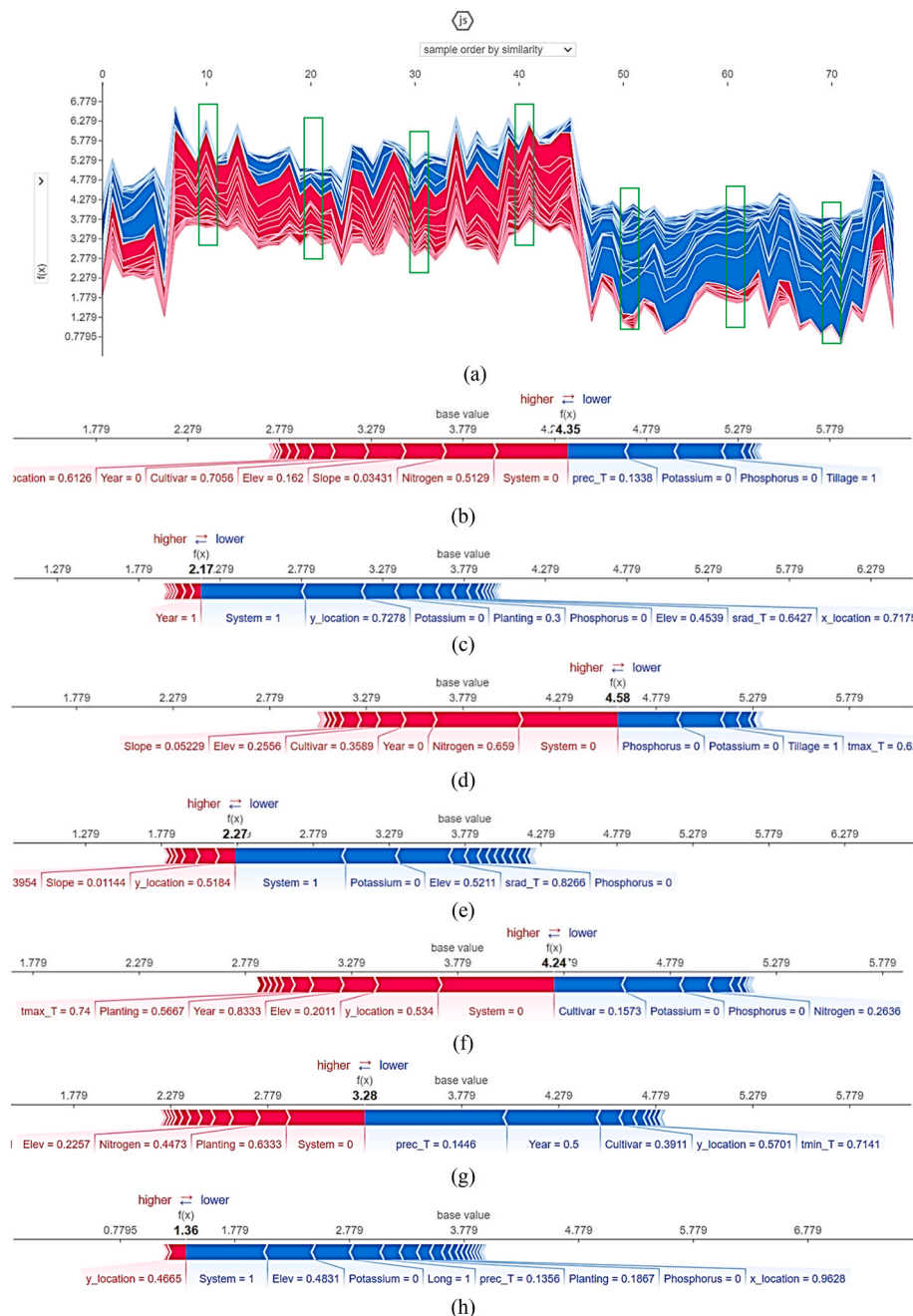


Fig. 19. Visualization of maize yield force plot: (a) stacked plot depicting overall trends, and (b–h) individual plots showcasing diverse observations.

potassium, phosphorus, and tillage. The output value for the model's prediction shown in the individual force plot of Fig. 19 (a) is 4.35.

While the system parameter tends to have the most positive impact across multiple observations, other variables exhibit varying degrees of influence as the observation changes. However, it is important to note that the force plots for individual observations (Fig. 19(b–h)) reveal changes in the influential variables. Different observations exhibit variations in the parameters that have the most positive impact and those that have the most negative impact on the model's prediction. The system parameter may consistently appear as a positive influencer in most observations, but other variables change sides or differ in their impact.

For instance, at certain observation points, precipitation (Prcp_T) emerges as the variable with the most significant negative impact on the model's prediction, followed by potassium, phosphorus, and tillage. These observations highlight the dynamic nature of the variables and how their contributions can differ across different instances. The collective analysis of these individual force plots emphasizes the varying importance of parameters at different observation points and underscores the need to consider specific characteristics when predicting maize yield accurately.

The absolute summary Beeswarm plot is represented in Fig. 20. The Beeswarm plot is a powerful visualization technique that illustrates the importance of features by presenting a scatterplot-like representation. It provides valuable insights into the distribution of feature contributions, allowing for a clear understanding of the range and density of importance values. The horizontal positioning of the points ensures that they do not overlap, resulting in optimal clarity.

In a Beeswarm plot, each variable is represented by individual points corresponding to each instance (i.e., row) in the dataset. These points are horizontally distributed along the x-axis based on their corresponding SHAP value. When there is a high density of SHAP values, the points are stacked vertically, highlighting areas of concentrated

influence. By examining the distribution of SHAP values, we can gain insights into how a variable influences the model predictions.

Fig. 20 presents a visualization of the Beeswarm plot, showcasing the top 20 variables used in SHAP computation. An overview of the magnitude is shown by the colorful line on the right, which contains all the variables used in the model. The variables are ordered in descending order based on their mean absolute SHAP values, indicating their relative significance. System, location, nitrogen, potassium, elevation, cultivar, and precipitation have a high impact on model output while CEC, temperature, tillage, and clay has the lowest impact on model output. The chart illustrates how the ML model's prediction of maize production is influenced by varying values of different variables. Positive SHAP values indicate an increase in the projected model estimate of maize yield. It appears from the summary plot that the presence of the variable system (higher values shown in red on the horizontal bar) implies a decrease in maize yield predicted (expressed on the scale at the bottom of the figure); on the other hand, a decrease in maize yield predicted will result from the absence of the variable system. The same methodology may be applied to the remaining factors.

4. Discussion

This study tries to highlight the complex interactions between different crop management practices, ambient conditions, and different soil qualities in determining maize yield. Understand the underlying mechanisms and potential synergies between these factors is of utmost important for enhancing crop productivity.

The influence of soil pH on maize yield was insignificant within the pH range of 5–8. While pH is significant for soil fertility and nutrient availability, other factors such as precipitation, solar radiation, and temperature play a more dominant role in determining crop yield. It is possible that the interplay between these factors masks the direct influence of pH within this particular pH range. Adequate water availability is crucial for optimal maize yields, as observed in the relationship between precipitation and crop productivity within the range of 5–15 mm day⁻¹. Maize plants may experience water stress at lower precipitation levels, leading to reduced growth and yield. Excessive precipitation, on the other hand, can result in waterlogging, affecting root health and nutrient uptake. The optimal precipitation range of 5–15 mm day⁻¹ provides the right balance for maximizing maize yield [12,14].

Solar radiation had a significant impact on maize yields. Values ranging from 16 to 20 MJ m⁻² day⁻¹ were associated with the highest yields. Solar radiation is vital for photosynthesis and energy production in plants. Higher solar radiation levels provide more energy for plant growth and development, ultimately leading to increased crop yields. Therefore, considering local climatic conditions and ensuring sufficient exposure to solar radiation are essential for optimizing maize production [57,58]. However, our study concentrated on the relationship between solar radiation and maize yield under the specific environmental conditions of the studied region. We did not delve into the effects of water stress on different maize cultivars or examine variables like radiation use efficiency (RUE) and water use efficiency (WUE) under varied water stress conditions. Hence, our observations on solar radiation's positive impact pertain to the non-water-stressed conditions prevalent during our study period.

While no clear linear trend was observed between soil pH and maximum and minimum temperatures, specific temperature and vapor pressure ranges showed a more favorable trend for higher crop yields [59]. Temperature ranges of 25–35 °C and vapor pressure ranges of 1500–2000 Pa promote optimal plant physiological processes, including photosynthesis, nutrient uptake, and growth. Vapor pressure, which affects plant transpiration and water movement, also influences nutrient uptake [60]. These identified temperature and vapor pressure ranges can be considered as targets for optimizing maize yield.

Moreover, the maize crop yield is greatly influenced by various farm management practices, including system selection, tillage practices

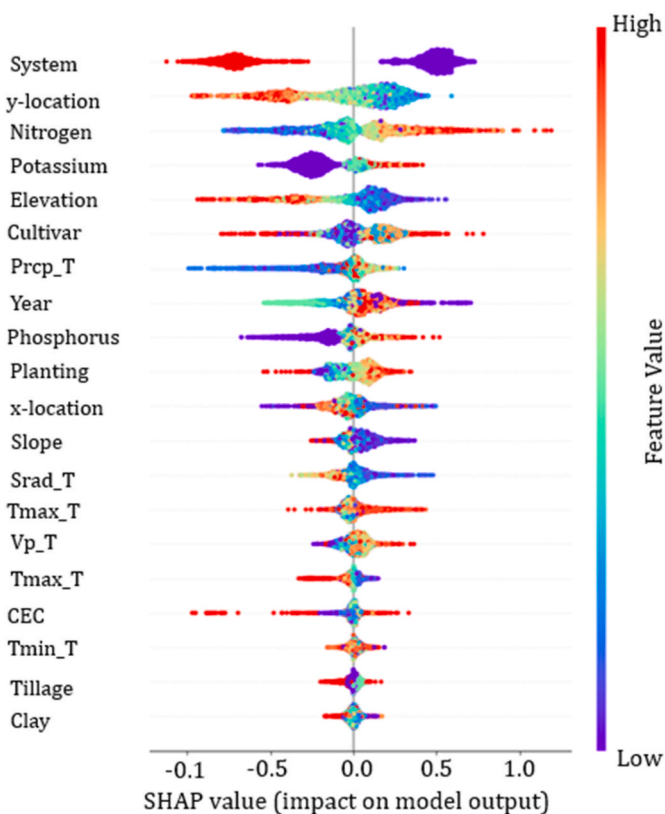


Fig. 20. Impact summary plot illustrating the influence of various features on the maize yield output (YIELD).

[19], choice of cultivar, and appropriate planting dates. These factors play a significant role in determining the overall productivity of the crop. The study findings suggest that the implementation of mechanized and medium-input techniques such as the Hybrid system, has a substantial positive impact on maize yield compared to Landrace-based system. The hybrid system, benefiting from mechanization and medium-input resources, achieves much higher yields (4.481 Mg ha^{-1}) compared to Landrace (1.160 Mg ha^{-1}) [61]. The adoption of advanced techniques and technologies, along with the focus on commercial production and local market selling, incentivizes farmers to maximize their yield potential. In contrast, the Landrace-based system, representing small-scale farmers focused on self-consumption, operates with limited resources and rain-fed farming practices, resulting in lower yields.

Furthermore, our data analysis indicates that tillage practices significantly affect maize yield. Reduced tillage yields the highest, marginally surpassing both no-tillage and conventional tillage. The results suggest that reduced tillage, which combines elements of both conventional and no-tillage practices, can be highly effective in the specific environmental and agricultural context of Chiapas, Mexico [19]. This approach seems to offer a balanced mix of soil disturbance and preservation. This may contribute to optimal conditions for maize growth and yield realization in this region.

No-tillage, yielding slightly less than reduced tillage but more than conventional tillage, continues to be a viable option. It preserves soil structure, moisture, and nutrients, promoting sustainable and productive crop production. The small yield difference between reduced tillage and no-tillage indicates that both practices can be effective alternatives to conventional tillage. Conventional tillage, while still a common practice, exhibits the lowest yield among the three methods in our study. Although it has been traditionally associated with higher yields, our findings suggest that in certain contexts, like Chiapas, alternative tillage methods might be more beneficial. Conventional tillage might also pose challenges such as soil erosion, loss of soil organic matter, and increased energy consumption. Based on these findings, it is imperative to consider regional specificities when deciding on the most appropriate tillage practice. Our study highlights the potential benefits of reduced and no-tillage practices in terms of yield and sustainability, particularly in the context of smallholder farming in tropical regions like Chiapas [62].

When choosing cultivars and varieties, it is essential to consider factors such as yield, quality, and agronomic characteristics. Among these, yield potential and adaptability play crucial roles in maize varieties. Maize varieties with high yield potential produce abundant harvests, while adaptability refers to their ability to thrive under various environmental conditions [63]. This adaptability is especially significant as it enables a variety to withstand challenges such as drought, heat, and cold. This ensures its resilience and productivity in diverse situations.

Additionally, the planting seed dates plays a crucial role in maize crop success and yield. Optimal planting timing allows farmers to take advantage of favorable conditions, such as temperature, daylight hours, and water availability. It helps mitigate risks associated with temperature fluctuations, frost, pests, and diseases. Selecting the appropriate seed date ensures timely maturity during the growing season, facilitating timely harvests and minimizing yield losses [63].

The impact of slope and fertilization on maize yield is of significant importance in agriculture research. Understanding how these factors influence crop productivity is crucial for optimizing cultivation strategies and maximizing grain production. The findings in this study indicate that concentrating maize production in the footslope position yields higher grain yields than in other slope positions. It is likely that the result is due to soil leaching and nutrient loss in the upper slope positions. Runoff from the surface and interflow from the subsurface are responsible for this. Furthermore, lower slopes tend to have deeper soil profiles, creating better moisture conditions for maize [64].

The impact of fertilization practices on maize yield is also worth considering. Adequate fertilization can enhance nutrient availability,

promote plant growth, and ultimately increase grain yields [17]. However, it is imperative to note that excessive fertilizer application may lead to environmental pollution and nutrient imbalances. Finding the right balance and optimizing fertilization practices is crucial for sustainable agriculture. Future research could delve deeper into determining the optimal fertilizer rates and timing specific to different slope positions to further enhance maize productivity [40].

The findings underscore the critical role of various factors, including crop and farm management practices, ambient conditions, and soil quality parameters, in influencing maize yields. Considering these factors becomes paramount when implementing effective maize cultivation strategies, as they can substantially impact grain yields. By gaining a comprehensive understanding of the relationships between these parameters and maize productivity, farmers can make informed decisions to maximize their yield potential. It is essential to prioritize sustainable and environmentally friendly agricultural practices to ensure long-term viability and ecological balance in maize cultivation.

5. Conclusion

In this study, we conducted a comprehensive analysis using extensive data analytics techniques. We investigated the impact of various factors on maize yield in smallholder farming in tropical regions. Our findings revealed significant associations between maize yield and a range of variables, including climate factors, soil quality parameters, cultivars, tillage practices, fertilizer usage, and notably, solar radiation.

Climate Variables and Maize Yield: Our analysis demonstrated a clear correlation between climate variables and maize yield. Particularly, solar radiation emerged as a critical factor with a strong positive correlation to maize production. Increased solar radiation positively impacts crop growth and yield, highlighting its pivotal role in tropical maize farming. Understanding these climate-driven trends is crucial for farmers and policymakers to harness solar radiation and adapt to changing weather patterns.

Soil Quality Parameters: Soil quality also determines crop productivity. The levels of nutrients, organic matter, and pH in the soil were significant factors. Implementing soil improvement strategies based on these findings can enhance agricultural practices and maximize maize yields in conjunction with favorable solar radiation.

Cultivars and Tillage Practices: The choice of maize cultivars and tillage practices have been among the most influential factors. Some cultivars are known to be better suited to specific local conditions, and various tillage techniques have been documented to enhance soil structure and moisture retention. These insights can guide farmers in selecting the most suitable cultivars and practices for their specific contexts.

Fertilizer Usage: Fertilizer application practices were another key determinant of maize yield, with a need for balanced nutrient management, focusing on nitrogen and potassium. This knowledge can lead to more efficient fertilizer use, reducing costs and environmental impacts while maximizing solar radiation synergy.

By employing data visualization and analytics, we uncovered spatial and temporal patterns, providing valuable insights into maize production factors. Furthermore, we utilized machine learning techniques to develop predictive models for maize yield. After comparing six different models, we determined that the grid search-based XGBoost model outperformed the others in terms of accuracy and reliability. This optimized model serves as a valuable tool for informed crop management decision-making.

Our data analytics and machine learning approach led to the development of a predictive maize yield model. The Grid search-based XGBoost model outperforms others in accuracy and reliability, providing a valuable tool for informed decision-making in crop management practices. Incorporating explainable artificial intelligence (XAI), we identified key contributing factors. Notably, solar radiation, system parameters, nitrogen application, slope, elevation, cultivar, and

precipitation had high impacts on model outputs, while other factors had lower impacts. Our study underscores the multifaceted nature of factors influencing maize yield in smallholder farming in tropical regions. Solar radiation, in particular, stands out as a critical driver of maize production with its strong positive correlation. The combination of data analytics, machine learning, and XAI not only advances our understanding of these factors but also provides practical tools for stakeholders to drive sustainable agricultural practices. This will enable stakeholders to harness solar radiation, and bolster food security in these critical regions. These findings represent a significant step towards resilient and productive farming systems in the face of evolving environmental challenges. We acknowledge the limitations of our study in terms of its regional focus and have suggested that future research should include similar studies in other tropical locations.

CRediT authorship contribution statement

Showkat Ahmad Bhat: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Syed Asif Ahmad Qadri:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis. **Vijay Dubbey:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation. **Ishfaq Bashir Sofi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Nen-Fu Huang:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This work is supported by the National Science and Technology Council (MOST 111-2221-E-007-063-MY3 and MOST 111-2622-8-007-008-TE2) of Taiwan.

References

- [1] B. Jama, et al., Maize yield response, nitrogen use efficiency and financial returns to fertilizer on smallholder farms in southern Africa, *Food Secur.* 9 (3) (2017) 577–593.
- [2] C. Ureta, et al., Maize yield in Mexico under climate change, *Agric. Syst.* 177 (2020) 102697.
- [3] M. Asim, et al., Responses of maize to different levels and sources of phosphorus application, *Pure Appl. Biol. (PAB)* 6 (3) (2017) 1030–1036.
- [4] M. Obadi, Y. Qi, B. Xu, High-amylose maize starch: structure, properties, modifications and industrial applications, *Carbohydr. Polym.* (2022) 120185.
- [5] GYGA, Global yield gap atlas [cited 2023 05/15], <https://www.yieldgap.org/gv/gviewer/index.html>, 2021.
- [6] F. Stat, FAO Statistical Year Book, Food and Agriculture organization of the United Nations, Rome, Italy, 2016.
- [7] ASERCA, Maíz grano cultivo representativo de México. Agencia de Servicios a la Comercialización y Desarrollo de Mercados Agropecuarios, 2018 [cited 2023 05/15], <https://www.gob.mx/aserca/articulos/maiz-grano-cultivo-representativo-demexico?idiom=es>.
- [8] FIRA, Dirección de Investigación y Evaluación Económica y Sectorial. Panorama Agroalimentario. Fideicomisos Instituidos en Relación con la Agricultura, México, 2016, 2023, <https://www.gob.mx/fira/documentos/panoramaagroalimentario>.
- [9] C.B. Field, V.R. Barros, Climate Change 2014–Impacts, Adaptation and Vulnerability: Global and Sectoral Aspects, Cambridge University Press, 2014.
- [10] O. Erenstein, et al., Global maize production, consumption and trade: trends and R&D implications, *Food Secur.* 14 (5) (2022) 1295–1319.
- [11] M. Ahmadi, H.R. Etedali, A. Elbeltagi, Evaluation of the effect of climate change on maize water footprint under RCPs scenarios in Qazvin plain, Iran, *Agric. Water Manag.* 254 (2021) 106969.
- [12] Y. Zou, et al., Deficit irrigation improves maize yield and water use efficiency in a semi-arid environment, *Agric. Water Manag.* 243 (2021) 106483.
- [13] P. Zhu, et al., Dissecting the nonlinear response of maize yield to high temperature stress with model-data integration, *Global Change Biol.* 25 (7) (2019) 2470–2484.
- [14] E.M. Adamgbe, F. Ujoh, Effect of Variability in Rainfall Characteristics on Maize Yield in Gboko, 2013. Nigeria.
- [15] X. Yan, et al., What are the key factors affecting maize yield response to and agronomic efficiency of phosphorus fertilizer in China? *Field Crops Res.* 270 (2021) 108221.
- [16] S. Qiang, et al., Maize yield, rainwater and nitrogen use efficiency as affected by maize genotypes and nitrogen rates on the Loess Plateau of China, *Agric. Water Manag.* 213 (2019) 996–1003.
- [17] I. Bekele, et al., Response of maize yield to nitrogen, phosphorus, potassium and sulphur rates on Andosols and Nitisols in Ethiopia, *Exp. Agric.* 58 (2022).
- [18] H. Ngoma, N.M. Mason, N.J. Sitko, Does minimum tillage with planting basins or ripping raise maize yields? Meso-panel data evidence from Zambia, *Agric. Ecosyst. Environ.* 212 (2015) 21–29.
- [19] M. Githongo, et al., The effect of minimum tillage and animal manure on maize yields and soil organic carbon in sub-Saharan Africa: a meta-analysis, *Environ. Chall.* 5 (2021) 100340.
- [20] M. Croci, et al., Dynamic maize yield predictions using machine learning on multi-source data, *Rem. Sens.* 15 (1) (2022) 100.
- [21] S.A. Bhat, N.-F. Huang, Big data and ai revolution in precision agriculture: survey and challenges, *IEEE Access* 9 (2021) 110209–110222.
- [22] B. Basso, L. Liu, Seasonal crop yield forecast: methods, applications, and accuracies, *Adv. Agron.* 154 (2019) 201–255.
- [23] Y. Ma, et al., Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach, *Rem. Sens. Environ.* 259 (2021) 112408.
- [24] S. Khaki, H. Pham, L. Wang, Simultaneous corn and soybean yield prediction from remote sensing data using deep transfer learning, *Sci. Rep.* 11 (1) (2021) 11132.
- [25] Y. Li, et al., Toward building a transparent statistical model for improving crop yield prediction: modeling rainfed corn in the US, *Field Crops Res.* 234 (2019) 55–65.
- [26] Y. Kang, et al., Comparative assessment of environmental variables and machine learning algorithms for maize yield prediction in the US Midwest, *Environ. Res. Lett.* 15 (6) (2020) 064005.
- [27] D. Lee, et al., Maize yield forecasts for Sub-Saharan Africa using Earth Observation data and machine learning, *Global Food Secur.* 33 (2022) 100643.
- [28] Y. Ren, et al., Analysis of corn yield prediction potential at various growth phases using a process-based model and deep learning, *Plants* 12 (3) (2023) 446.
- [29] P. Jayagopal, et al., Weather-based maize yield forecast in Saudi Arabia using statistical analysis and machine learning, *Acta Geophys.* 70 (6) (2022) 2901–2916.
- [30] C. Ureta, et al., Projecting the effects of climate change on the distribution of maize races and their wild relatives in Mexico, *Global Change Biol.* 18 (3) (2012) 1073–1082, 2012.
- [31] Ray, D., Gerber, J., MacDonald, G. et al., Climate variation explains a third of global crop yield variability. *Nat. Commun.* 6 (5989).
- [32] C. Ureta, et al., Finding potential high-yield areas for Mexican maize under current and climate change conditions, *J. Agric. Sci.* 154 (5) (2016) 782–794.
- [33] A. Gardezabal, et al., Knowledge management for innovation in agri-food systems: a conceptual framework, *Knowl. Manag. Res. Pract.* 21 (2) (2023) 303–315.
- [34] SIAP, Estadística de Producción Agrícola. 2020]. <http://infosiap.siap.gob.mx/gobmx/datosAbiertos.php>, 2020.
- [35] H. Wickham, et al., Package ‘dplyr’. A grammar of data manipulation, R Packag. version 8 (2019).
- [36] R.J. Hijmans, et al., Package ‘raster’, R Packag. 734 (2015) 473.
- [37] E. Luedeling, S. Siebert, A. Buerkert, Filling the voids in the SRTM elevation model—a TIN-based delta surface approach, *ISPRS J. Photogrammetry Remote Sens.* 62 (4) (2007) 283–294.
- [38] P.E. Thornton, et al., Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 2, Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States), 2014.
- [39] R.G. Trevisan, et al., Multiyear maize management dataset collected in Chiapas, Mexico, Data Brief 40 (2022) 107837.
- [40] L.E. Nemadodzi, et al., Nitrogen, phosphorus, and potassium effects on the physiology and biomass yield of baby spinach (*Spinacia oleracea* L.), *J. Plant Nutr.* 40 (14) (2017) 2033–2044.
- [41] J.D. Munoz, et al., Cover crop effect on corn growth and yield as influenced by topography, *Agric. Ecosyst. Environ.* 189 (2014) 229–239.
- [42] J. Kumhálová, et al., The impact of topography on soil properties and yield and the effects of weather conditions, *Precis. Agric.* 12 (2011) 813–830.
- [43] S. Machado, et al., Spatial and temporal variability of corn growth and grain yield: implications for site-specific farming, *Crop Sci.* 42 (5) (2002) 1564–1576.
- [44] S.A. Bhat, et al., Correlating the ambient conditions and performance indicators of the LoRaWAN via surrogate Gaussian process based bidirectional LSTM stacked autoencoder showkat, *IEEE Trans. Netw. Serv. Manag.* 20 (3) (2023) 3413–3427.
- [45] M. Shahhosseini, et al., Coupling machine learning and crop modeling improves crop yield prediction in the US Corn Belt, *Sci. Rep.* 11 (1) (2021) 1–15.
- [46] Z. Xiong, et al., Evaluating explorative prediction power of machine learning algorithms for materials discovery using k-fold forward cross-validation, *Comput. Mater. Sci.* 171 (2020) 109203.

- [47] P. Liashchynskiy, P. Liashchynskiy, Grid search, random search, genetic algorithm: a big comparison for NAS, arXiv preprint arXiv:1912.06059 (2019) 1–11.
- [48] R.J. Chase, et al., A machine learning tutorial for operational meteorology. Part I: traditional machine learning, *Weather Forecast.* 37 (8) (2022) 1509–1529.
- [49] S.A. Bhat, I. Hussain, N.-F. Huang, Soil suitability classification for crop selection in precision agriculture using GBRT-based hybrid DNN surrogate models, *Ecol. Inf.* (2023) 102109.
- [50] O. Lyashevskaya, et al., Long-term trends in herring growth primarily linked to temperature by gradient boosting regression trees, *Ecol. Inf.* 60 (2020) 101154.
- [51] R.T.-C. Sheng, et al., Rice growth stage classification via RF-based machine learning and image processing, *Agriculture* 12 (12) (2022) 2137.
- [52] R. Muñoz-Mas, et al., Tree-based ensembles unveil the microhabitat suitability for the invasive bleak (*Alburnus alburnus* L.) and pumpkinseed (*Lepomis gibbosus* L.): introducing XGBoost to eco-informatics, *Ecol. Inf.* 53 (2019) 100974.
- [53] X. Sun, M. Liu, Z. Sima, A novel cryptocurrency price trend forecasting model based on LightGBM, *Finance Res. Lett.* 32 (2020) 101084.
- [54] T. Hastie, et al., Boosting and additive trees, in: *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2009, pp. 337–387.
- [55] L. Breiman, Stacked regressions, *Mach. Learn.* 24 (1996) 49–64.
- [56] M.-C. Chiu, et al., Development of smart aquaculture farm management system using IoT and AI-based surrogate models, *J. Agric. Food Res.* 9 (2022) 100357.
- [57] R.C. Muchow, T.R. Sinclair, J.M. Bennett, Temperature and solar radiation effects on potential maize yield across locations, *Agron. J.* 82 (2) (1990) 338–343.
- [58] F. Andrade, S. Uhart, A. Cirilo, Temperature affects radiation use efficiency in maize, *Field Crops Res.* 32 (1–2) (1993) 17–25.
- [59] N. Sirisuntornlak, et al., Interactive effects of silicon and soil pH on growth, yield and nutrient uptake of maize, *Silicon* 13 (2) (2021) 289–299.
- [60] E. Li, et al., The compound effects of drought and high temperature stresses will be the main constraints on maize yield in Northeast China, *Sci. Total Environ.* 812 (2022) 152461.
- [61] M. Sanaullah, et al., Terrestrial ecosystem functioning affected by agricultural management systems: a review, *Soil Tillage Res.* 196 (2020) 104464.
- [62] K. Romanekas, et al., The impact of five long-term contrasting tillage systems on maize productivity parameters, *Agric. Food Sci.* 29 (1) (2020) 6–17, 6–17.
- [63] H. Mugiyi, et al., Investigation of the optimum planting dates for maize varieties using a hybrid approach: a case of Hwedza, Zimbabwe, *Heliyon* 7 (2) (2021) e06109.
- [64] K. Wolka, et al., Soil and water conservation management on hill slopes in southwest Ethiopia. II. Modeling effects of soil bunds on surface runoff and maize yield using AquaCrop, *J. Environ. Manag.* 296 (2021) 113187.